columns, each of which is a 3-tuple.

b. A matrix of size m by n stored in row-major form.

c. A three-dimensional array of size l by m by n stored as an l-tuple where each component is an m by n matrix stored in row-major form.

1.4 Graphs and Trees

When we think about graphs we might think about pictures of some kind that are used to represent information. The graphs that we'll discuss can be thought about in the same way. But we need to describe them in a little more detail if they are to be much use to us. We'll also see that trees are special kinds of graphs.

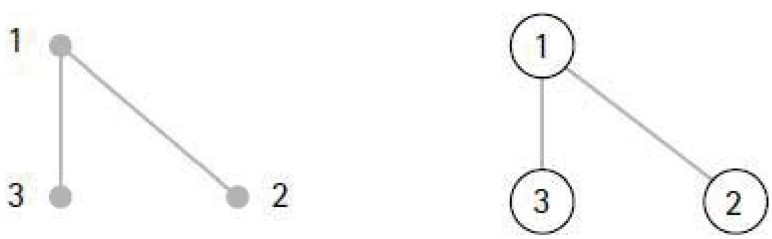


Figure 1.11 Graphs.

1.4.1 Definition of a Graph

Informally, a *graph* is a set of objects in which some of the objects are connected to each other in some way. The objects are called *vertices* or *nodes*, and the connections are called *edges*. For example, the United States can be represented by a graph where the vertices are states and the edges are the common borders between adjacent states. In this case, Hawaii and Alaska would be vertices without any edges connected to them. We say that two vertices are *adjacent* if there is an edge connecting them.

Picturing a Graph

We can picture a graph in several ways. For example, Figure 1.11 shows two ways to represent the graph with vertices 1, 2, and 3 and edges connecting 1 to 2 and 1 to 3.

example 1.34 States and Provinces

Figure 1.12 represents a graph, where the vertices are those states in the United States and those provinces in Canada that border the Pacific Ocean or that border states and provinces that touch the Pacific Ocean, and where an edge denotes a common border.

end example

Coloring a Graph

An interesting problem dealing with maps is to try to color a map with the fewest number of colors subject to the restriction that any two adjacent areas must have distinct colors. From a graph point of view, this means that any two distinct adjacent vertices must have different colors. Before reading any further, try to color the graph in Figure 1.12 with the fewest colors. It's usually easier to represent the colors by numbers like 1, 2, ...

A graph is *n-colorable* if there is an assignment of n colors to its vertices such that any two distinct adjacent vertices have distinct colors. The *chromatic number* of a graph is the smallest n for which it is n-colorable. For example, the chromatic number of the graph in Figure 1.12 is 3. A graph whose edges are the connections between all pairs of distinct vertices is called a *complete graph*. It's easy to see that the chromatic number of a complete graph with n vertices is n.

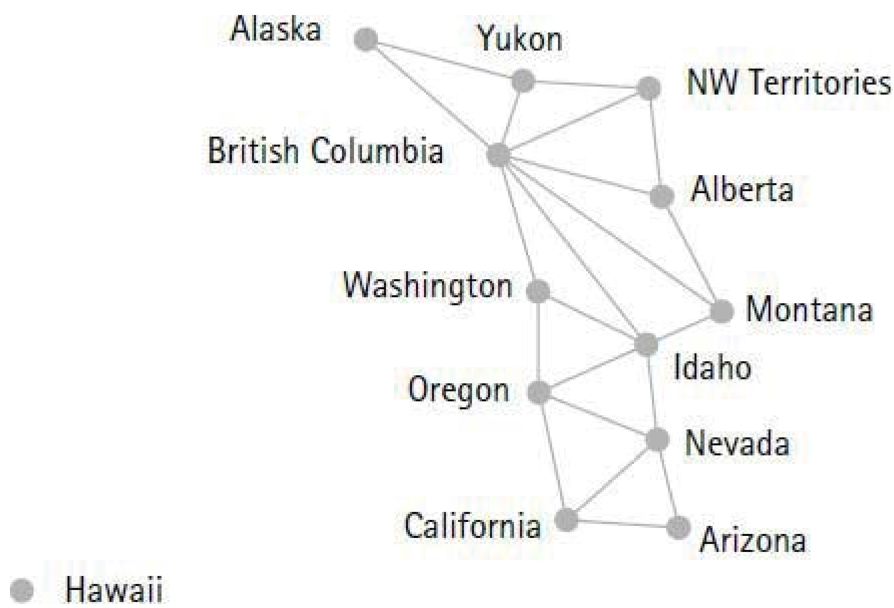


Figure 1.12 Graph of states and provinces.

A graph is *planar* if it can be drawn on a plane such that no edges intersect. For example, the graph in Figure 1.12 is planar. A complete graph with four vertices is planar, but a complete graph with five vertices is not planar. See whether you can convince yourself of these facts. A fundamental result on graph coloring—which remained an unproven conjecture for over 100 years—states that *every planar graph is 4-colorable*. The result was proven in 1976 by Kenneth Appel and Wolfgang Haken [1976, 1977]. They used a computer to test over 1900 special cases.

More Terminology

A *directed graph* (*digraph* for short) is a graph where each edge points in one direction. For example, the vertices could be cities and the edges could be the one-way air routes between them. For digraphs we use arrows to denote the edges. For example, Figure 1.13 shows two ways to represent the digraph with three vertices a, b, and c and edges from a to b, c to a, and c to b.

The *degree* of a vertex is the number of edges that it touches. However, we add two to the degree of a vertex if it has a *loop*, which is an edge that starts and ends at the same vertex. For directed graphs the *indegree* of a vertex is the number of edges pointing at the vertex, whereas the *outdegree* of a vertex is the number of edges pointing away from the vertex. In a digraph a vertex is called a *source* if its indegree is zero and a *sink* if its outdegree is zero. For example, in the digraph of Figure 1.13, c is a source and b is a sink.

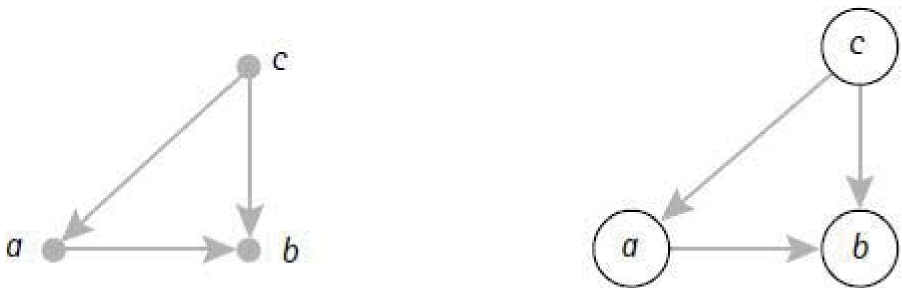


Figure 1.13 Directed graphs.

If a graph has more than one edge between some pair of vertices, the graph is called a *multigraph*, or a *directed multigraph* in case the edges point in the same direction. For example, there are usually two or more road routes between most cities. So a graph representing road routes between a set of cites is most likely a multigraph.

Representaions of Graphs

From a computational point of view, we need to represent graphs as data. This is easy to do because we can define a graph in terms of tuples, sets, and bags. For example, we can define a graph G as an ordered pair (V, E), where V is a set of vertices and E is a set orbag of edges. If G is a digraph, then the edges in E can be represented by ordered pairs, where (a, b) represents the edge with an arrow from a to b. In this case the set E of edges is a subset of $V \times V$. In other words, E is a binary relation on V. For example, the digraph in Figure 1.13 has vertex set $\{a, b, c\}$ and edge set

$$\{(a, b), (c, b), (c, a)\}.$$

If G is a directed multigraph, then we can represent the edges as a bag (or multiset) of ordered pairs. For example, the bag $[(a, b), (a, b), (b, a)]$ represents three edges: two from a to b and one from b to a.

If a graph is not directed, we have more ways to represent the edges. We could still represent an edge as an ordered pair (a, b) and agree that it represents an undirected line between a and b. But we can also represent an edge between vertices a and b by a set $\{a, b\}$. For example, the graph in

Figure 1.11 has vertex set $\{1, 2, 3\}$ and edge set $\{\{1, 2\}, \{1, 3\}\}$.

Weighted Graphs

We often encounter graphs that have information attached to each edge. For example, a good road map places distances along the roads between major intersections. A graph is called *weighted* if each edge is assigned a number, called a *weight*. We can represent an edge (a, b) that has weight w by the 3-tuple

$$(a, b, w).$$

In some cases we might want to represent an unweighted graph as a weighted graph. For example, if we have a multigraph in which we wish to distinguish between multiple edges that occur between two vertices, then we can assign a different weight to each edge, thereby creating a weighted multigraph.

Graphs and Binary Relations

We can observe from our discussion of graphs that any binary relation R on a set A can be thought of as a digraph $G = (A, R)$ with vertices A and edges R. For example, let $A = \{1, 2, 3\}$ and

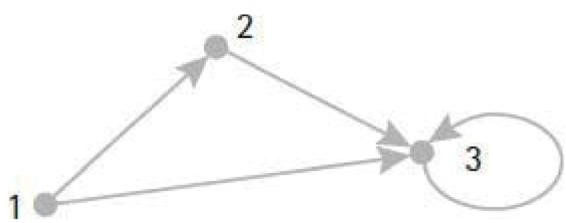


Figure 1.14 Digraph of binary relation.

$$R = \{(1, 2), (1, 3), (2, 3), (3, 3)\}.$$

Figure 1.14 shows the digraph corresponding to this binary relation. Representing a binary relation as a graph is often quite useful in trying to establish properties of the relation.

Subgraphs

Sometimes we need to discuss graphs that are part of other graphs. A graph (V', E') is a *subgraph* of a graph (V, E) if $V' \subseteq V$ and $E' \subseteq E$. For example, the four graphs in Figure 1.15 are subgraphs of the graph in Figure 1.14.

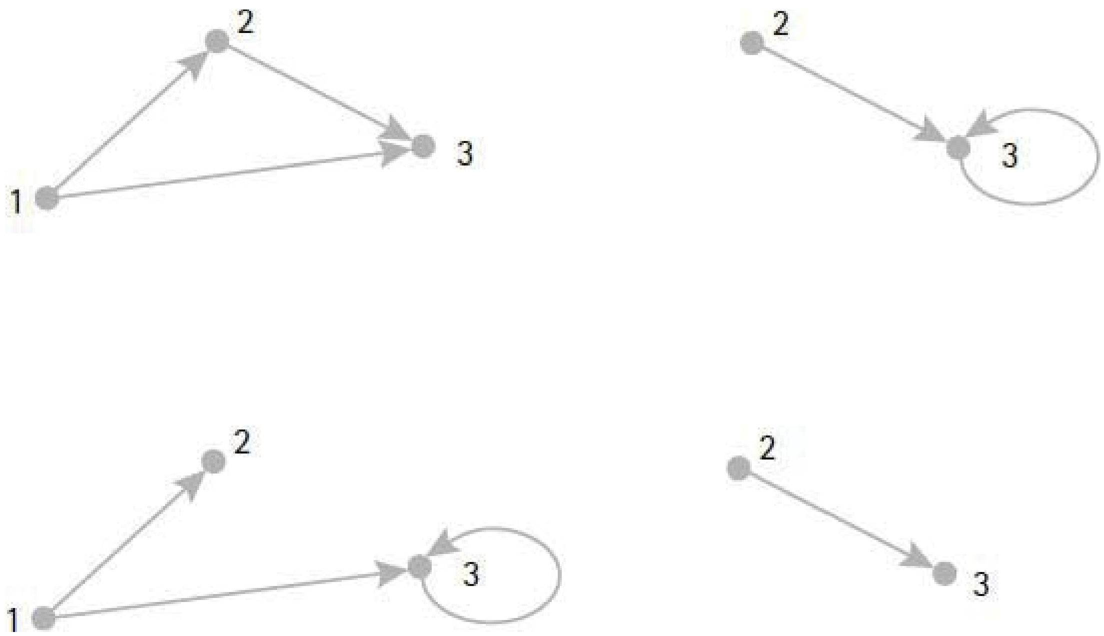


Figure 1.15 Subgraphs.

1.4.2 Paths in Graphs

Problems that use graphs often involve moving from one vertex to another along a sequence of edges, where each edge shares a vertex with the next edge in the sequence. In formal terms, a *path* from x_0 to x_n is a sequence of edges that we denote by a sequence of vertices $x_0, x_1, \ldots, x_n$ such that there is an edge from x_{i-1} to x_i for $1 \leq i \leq n$. A path allows the possibility that some edge or some vertex occurs more than once. A *cycle* is a path whose beginning and ending vertices are equal and in which no edge occurs more than once. A graph with no cycles is called *acyclic*. The *length* of the path $x_0, \ldots, x_n$ is the number n of edges.

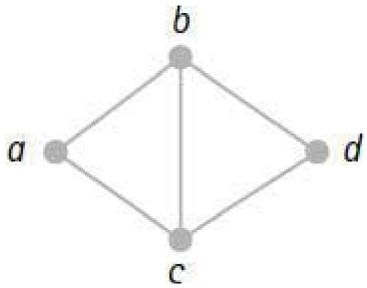


Figure 1.16 Sample graph.

Let's emphasize here that the definitions for path and cycle apply to both graphs and directed graphs.

example 1.35 Paths in a Graph

We'll examine a few paths in the graph pictured in Figure 1.16.

1. The path *b, c, d, b, a* visits *b* twice. The length of the path is 4.
2. The path *a, b, c, b, d* visits *b* twice and uses the edge between *b* and *c* twice. The length of the path is 4.
3. The path *a, b, c, a* is a cycle of length 3.
4. The path *a, b, a* is not a cycle because the edge from *a* to *b* occurs twice. The path has length 2.

end example

Connected Graphs

Now we come to an idea that needs a separate definition for each type of graph. A graph is *connected* if there is a path between every pair of vertices. For directed graphs there are two kinds of connectedness. A digraph is *weakly connected* if, when direction is ignored, the resulting undirected graph is connected. A digraph is *strongly connected* if there is a directed path between every pair of vertices. For example, the digraphs in Figure 1.13 and Figure 1.14 are weakly connected but not strongly connected.

Two Graph Problems

Let's look at two graph problems that you may have seen. For the first problem you are to trace the first diagram in Figure 1.17 without taking your pencil off the paper and without retracing any line.

After some fiddling, it's easy to see that it can be done by starting at one of the bottom two corners and finishing at the other bottom corner. The second diagram in Figure 1.17 emphasizes the graphical nature of the problem. From this point of view we can say that there is a path that travels each edge exactly once.

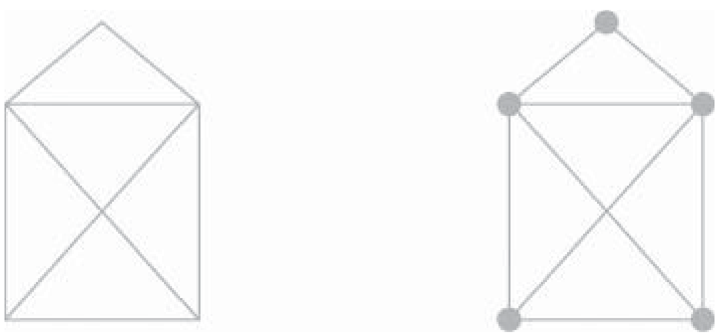


Figure 1.17 Tracing a graph.

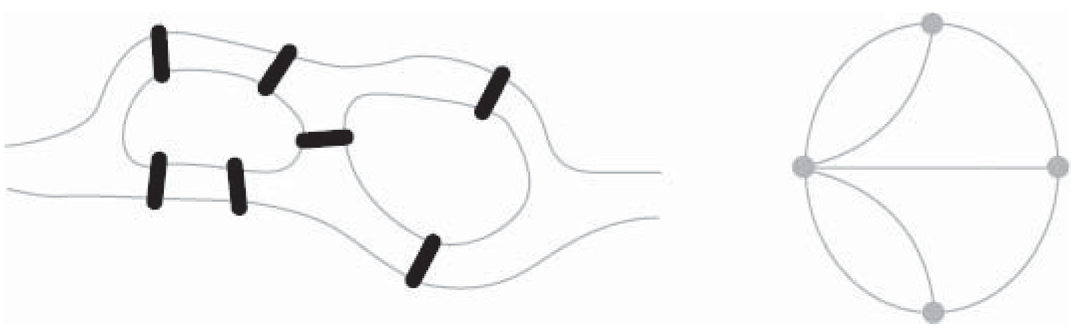


Figure 1.18 Seven bridges of Königsberg.

The second famous problem is named after the seven bridges of Königsberg that, in the early 1800s, connected two islands in the river Pregel to the rest of the town. The problem is to tour the town by walking across each of the seven bridges exactly once. In Figure 1.18 we've pictured the two islands and seven bridges of Königsberg together with a multigraph representing the situation. The vertices of the multigraph represent the four land areas and the edges represent the seven bridges.

The mathematician Leonhard Euler (1707–1783) proved that there aren't any such paths by finding a general condition for such paths to exist. In his honor, any path that contains each edge of a graph exactly once is called an *Euler path*. For example, the graph for the tracing problem has an Euler path, but the seven bridges problem does not. We can go one step further and define an *Euler circuit* to be any path that begins and ends at the same vertex and contains each edge of a graph exactly once. There are no Euler circuits in the graphs of Figure 1.17 and Figure 1.18. We'll discuss conditions for the existence of Euler paths and Euler circuits in the exercises.

1.4.3 Graph Traversals

A *graph traversal* starts at some vertex v and visits all vertices x that can be reached from v by traveling along some path from v to x. If a vertex has already been visited, it is not visited again. It follows that any traversal of a connected graph will visit all its vertices. Similarly, any traversal of a strongly connected digraph will visit all its vertices.

Let's look at two popular traversal algorithms: *breadth-first traversal* and *depth-first traversal*.

Breadth-First Traversal

To describe *breadth-first traversal*, we'll let visit(v, k) denote any procedure that visits every vertex x not yet visited for which there is a length k path from v to x. If the graph has n vertices, a breadth-first traversal starting at v can be described as follows:

$$\textbf{for } k := 0 \textbf{ to } n - 1 \textbf{ do } \text{visit}(v, k) \textbf{ od}.$$

Since we haven't specified how visit(v, k) does it's job, there are usually several different traversals from any given starting vertex.

example 1.36 Breadth-First Traversals

We'll do some breadth-first traversals of the graph in Figure 1.19. If we start at vertex a, then there are four possible breadth-first traversals, which we've represented by the following strings:

$$a\ b\ c\ d\ e\ f\ g \qquad a\ b\ c\ d\ f\ e\ g \qquad a\ c\ b\ d\ e\ f\ g \qquad a\ c\ b\ d\ f\ e\ g.$$

Of course, we can start a breadth-first traversal at any vertex. For example, one of several breadth-first traversals that start with *d* is represented by the string

$$d\ b\ e\ f\ a\ g\ c.$$

end example

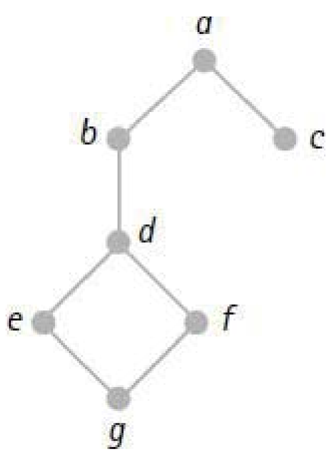


Figure 1.19 Sample graph.

Depth-First Traversal

We can describe *depth-first traversal* with a recursive procedure—one that calls itself. Let DF(v) denote the depth-first procedure that traverses the graph starting at vertex v. Then DF(v) has the following definition.

DF(v): **if** v has not been visited **then**
  visit v;
  for each edge from v to x **do** DF(x) **od**
fi

Since we haven't specified how each edge from v to x is picked in the **for** loop, there are usually several different traversals from any given starting vertex.

example **1.37 Depth-First Traversals**

We'll do some depth-first traversals of the graph represented in Figure 1.19. For example, starting from vertex a in the graph, there are four possible depth-first traversals, which are represented by the following strings:

$$a\ b\ d\ e\ g\ f\ c \qquad a\ b\ d\ f\ g\ e\ c \qquad a\ c\ b\ d\ e\ g\ f \qquad a\ c\ b\ d\ f\ g\ e.$$

end example

1.4.4 Trees

From an informal point of view, a *tree* is a structure that looks like a real tree. For example, a family tree and an organizational chart for a business are both trees. From a formal point of view we can say that a *tree* is a graph that is connected and has no cycles. Such a graph can be drawn to look like a real tree. The vertices and edges of a tree are called *nodes* and *branches*, respectively.

In computer science, and some other areas, trees are usually pictured as upside down versions of real trees, as in Figure 1.20. For trees represented this way, the node at the top is called the *root*. The nodes that hang immediately below a given node are its *children*, and the node immediately above a given node is its *parent*. If a node is childless, then it is a *leaf*. The *height* or *depth* of a tree is the length of the longest path from the root to the leaves. The *level* of a node is the length of the path from the root to the node. The path from a node to the root contains all the *ancestors* of the node. Any path from a node to a leaf contains *descendants* of the node.

A tree with a designated root is often called a *rooted tree*. Otherwise, it is called a *free tree* or an *unrooted tree*. We'll use the term *tree* and let the context indicate the type of tree.

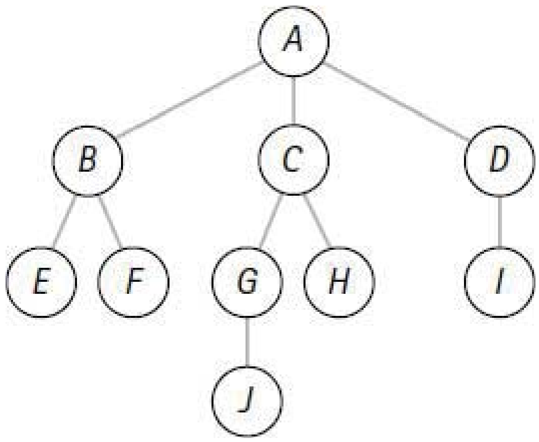


Figure 1.20 Sample tree.

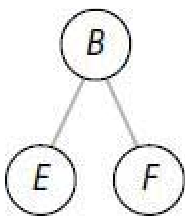


Figure 1.21 A subtree.

example 1.38 Parts of a Tree

We'll make some observations about the tree in Figure 1.20. The root is *A*. The children of *A* are *B, C*, and *D*. The parent of *F* is *B*. The leaves of the tree are *E, F, J, H*, and *I*. The height or depth of the tree is 3. The level of *H* is 2. The ancestors of *J* are *A, C*, and *G*. The descendants of *C* are *G, J*, and *H*.

end example

Subtrees

If *x* is a node in a tree *T*, then *x* together with all its descendants forms a tree *S* with *x* as its root. *S* is called a *subtree* of *T*. If *y* is the parent of *x*, then *S* is sometimes called a subtree of *y*.

example 1.39 A Subtree

The tree pictured in Figure 1.21 is a subtree of the tree in Figure 1.20. Since A is the parent of B, we can also say that this tree is a subtree of node A.

end example

Ordered and Unordered Trees

If we don't care about the ordering of the children of a tree, then the tree is called an *unordered tree*. A tree is *ordered* if there is a unique ordering of the children of each node. For example, any algebraic expression can be represented as an ordered tree.

example 1.40 Representing Algebraic Expressions

The expression $x - y$ can be represented by a tree whose root is the minus sign and with two subtrees, one for x on the left and one for y on the right. Ordering is important here because the subtraction operation is not commutative. For example, Figure 1.22 represents the expression $3 - (4 + 8)$ and Figure 1.23 represents the expression $(4 + 8) - 3$.

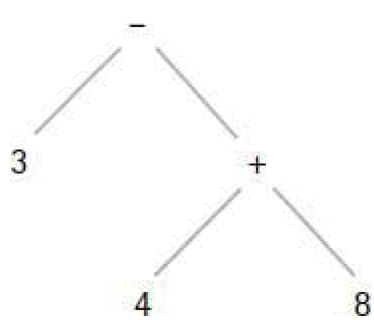


Figure 1.22 Tree for $3 - (4 + 8)$.

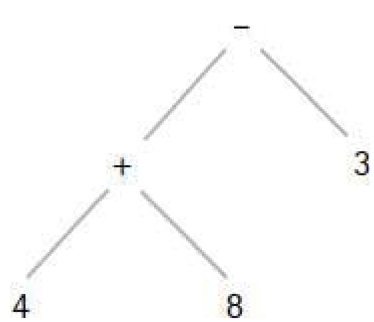


Figure 1.23 Tree for $(4 + 8) - 3$.

end example

Representations of Trees

How can we represent a tree as a data object? The key to any representation is that we should be able to recover the tree from its representation. One method is to let the tree be a list whose first element is the root and whose next elements are lists that represent the subtrees of the root.

For example, the tree with a single node r is represented by $\langle r$, and the list representation of the tree for the algebraic expression $a - b$ is

$$-, \ a \ , \ b \ .$$

For another example, the list representation of the tree for the arithmetic expression $3 - (4 + 8)$ is

$$-, \ 3 \ , \ +, \ 4 \ , \ 8 \ .$$

For a more complicated example, let's consider the tree represented by the following list.

$$T = \ r, \ b, \ c \ , \ d \ , \ x, \ y, \ z \ , \ w \ , \ e, \ u \ .$$

Notice that T has root r, which has the following three subtrees:

$$b, \ c \ , \ d$$

$$x, \ y, \ z \ , \ w$$

$$e, \ u \ .$$

Similarly, the subtree $b, \ c \ , \ d$ has root b, which has two children c and d. We can continue in this way to recover the picture of T in Figure 1.24.

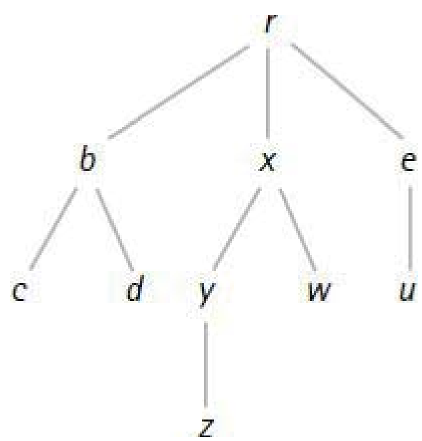
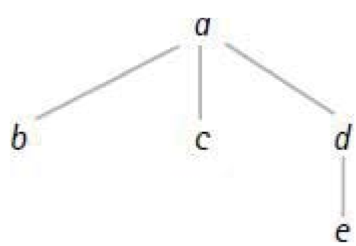


Figure 1.24 Sample tree.

example 1.41 Computer Representation of Trees

Let's represent a tree as a list and then see what it looks like in computer memory. For example, let T be the following tree.



We'll represent the tree as the list $T = \langle a, \langle b \rangle, \langle c \rangle, \langle d, \langle e \rangle \rangle \rangle$. Figure 1.25 shows the representation of T in computer memory, where we have used the same notation as Example 1.24.

end example

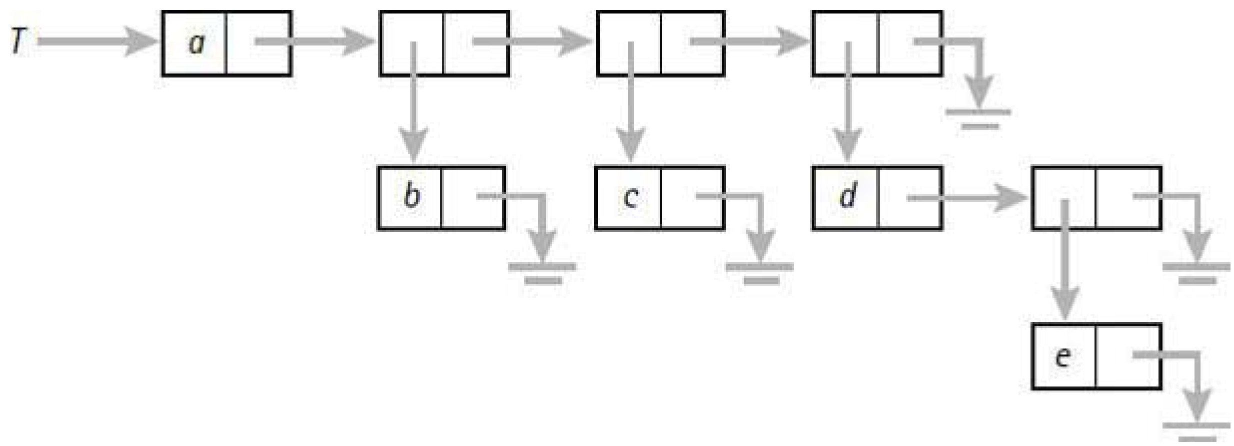


Figure 1.25 Computer representation of a tree.

Binary Trees

A *binary tree* is an ordered tree that may be empty or else has the property that each node has two subtrees, called the *left* and *right* subtrees of the node, which are binary trees. We can represent the empty binary tree by the empty list $\langle\ \rangle$. Since each node has two subtrees, we represent nonempty binary trees as 3-element lists of the form

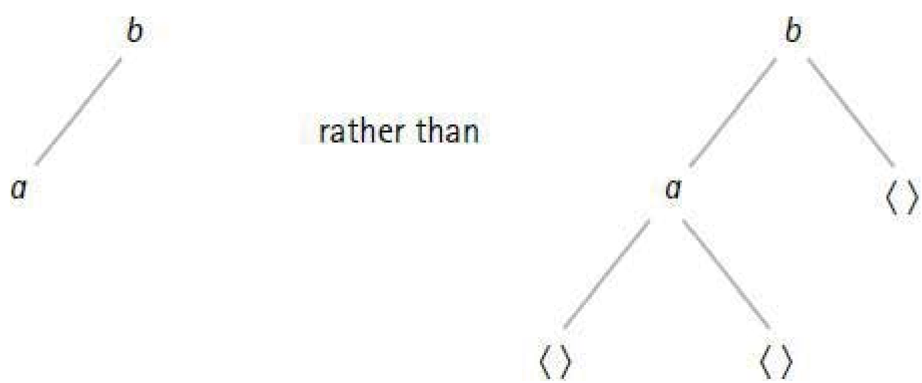


Figure 1.26 Simplified binary tree.

$$\langle L, x, R\rangle,$$

where x is the root, L is the left subtree, and R is the right subtree. For example, the tree with one node x is represented by the list $\langle\langle\rangle, x, \langle\rangle\rangle$.

When we draw a picture of a binary tree, it is common practice to omit the empty subtrees. For example, the binary tree represented by the list

$$\langle\langle\langle\rangle, a, \quad , b,$$

is usually, but not always, pictured as the simpler tree in Figure 1.26.

example 1.42 Computer Representation of Binary Trees

Let's see what the representation of a binary tree as a list looks like in computer memory. For example, let T be the following tree.

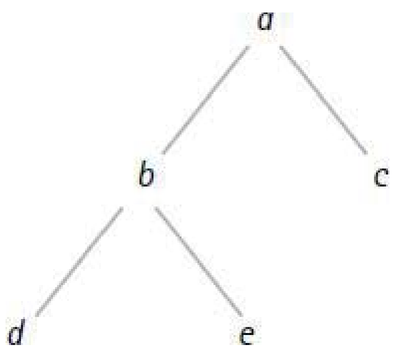


Figure 1.27 shows the representation of T in computer memory, where each block of memory contains a node and pointers to the left and right subtrees.

end example

Binary Search Trees

Binary trees can be used to represent sets whose elements have some ordering. Such a tree is called a *binary search tree* and has the property that for each node of the tree, each element in its left subtree precedes the node element and each element in its right subtree succeeds the node element.

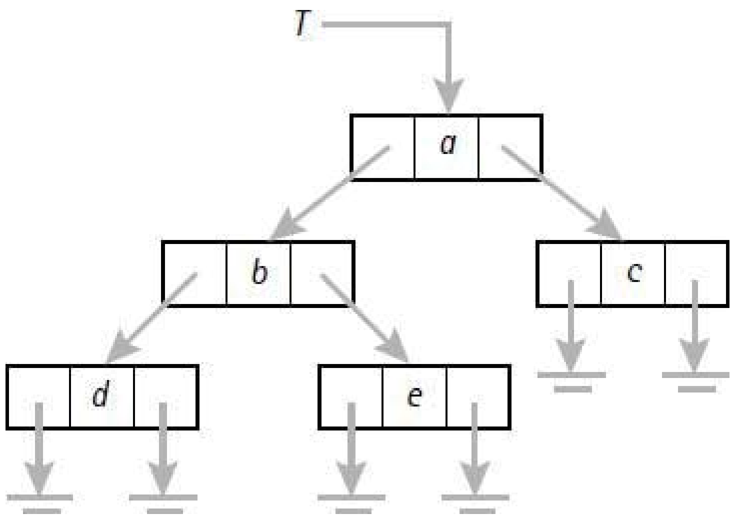


Figure 1.27 Computer representation of a binary tree.

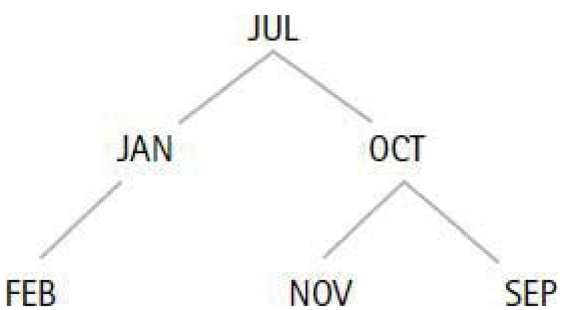


Figure 1.28 Binary search tree.

example 1.43 A Binary Search Tree

The binary search tree in Figure 1.28 holds three-letter abbreviations for six of the months, where we are using the dictionary ordering of the words. So the correct order is FEB, JAN, JUL, NOV, OCT, SEP.

This binary search tree has depth 2. There are many other binary search trees to hold these six months. Find another one that has depth 2. Then find one that has depth 3.

end example

1.4.5 Spanning Trees

A *spanning tree* for a connected graph is a subgraph that is a tree and contains all the vertices of the graph. For example, Figure 1.29 shows a graph followed by two of its spanning trees. This example shows that a graph can have many spanning trees. A *minimal spanning tree* for a connected weighted graph is a spanning tree such that the sum of the edge weights is minimum among all spanning trees.

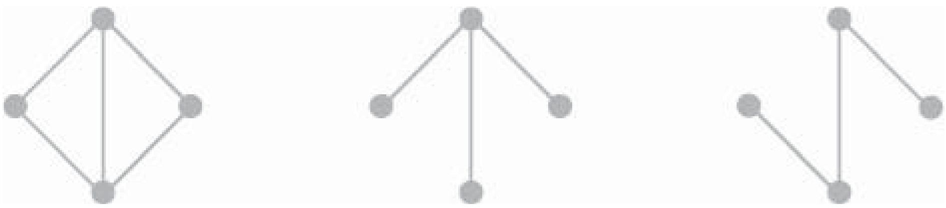


Figure 1.29 Two spanning trees.

Prim's Algorithm

A famous algorithm, due to Prim [1957], constructs a minimal spanning tree for any undirected connected weighted graph. Starting with any vertex, the algorithm searches for an edge of minimum weight connected to the vertex. It adds the edge to the tree and then continues by trying to find new edges of minimum weight such that one vertex is in the tree and the other vertex is not. Here's an informal description of the algorithm.

Prim's Algorithm

Construct a minimal spanning tree for an undirected connected weighted graph. The variables: V is the vertex set of the graph; W is the vertex set and S is the edge set of the spanning tree.

1. Initialize $S := \varnothing$.

2. Pick any vertex $v \in V$ and set $W := \{v\}$.

3. while $W \neq V$ **do**

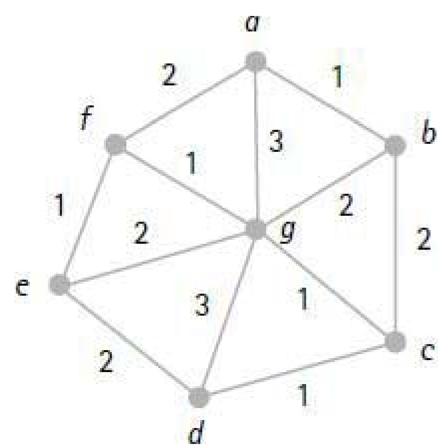
  Find a minimum weight edge $\{x, y\}$, where $x \in W$ and
   $y \in V - W$;
  $S := S \cup \{\{x, y\}\}$;
  $W := W \cup \{y\}$

od

Of course, Prim's algorithm can also be used to find a spanning tree for an unweighted graph. Just assign a weight of 1 to each edge of the graph. Or modify the first statement in the **while** loop to read "Find an edge $\{x, y\}$ such that $x \in W$ and $y \in V - W$."

example 1.44 A Minimal Spanning Tree

We'll construct a minimal spanning tree for the following weighted graph.



To see how the algorithm works we'll do a trace of each step showing the values of the variables S and W. The algorithm gives us several choices since it is not implemented as a computer program. So we'll start with the letter a since it's the first letter of the alphabet.

S	W
$\{\}$	$\{a\}$
$\{\{a,b\}\}$	$\{a,b\}$
$\{\{a,b\},\{b,c\}\}$	$\{a,b,c\}$
$\{\{a,b\},\{b,c\},\{c,d\}\}$	$\{a,b,c,d\}$
$\{\{a,b\},\{b,c\},\{c,d\},\{c,g\}\}$	$\{a,b,c,d,g\}$
$\{\{a,b\},\{b,c\},\{c,d\},\{c,g\},\{g,f\}\}$	$\{a,b,c,d,g,f\}$
$\{\{a,b\},\{b,c\},\{c,d\},\{c,g\},\{g,f\},\{f,e\}\}$	$\{a,b,c,d,g,f,e\}$

The algorithm stops because $W = V$. So S is a spanning tree.

end example

Exercises

Graphs

1. Draw a picture of a graph that represents those states of the United States and those provinces of Canada that touch the Atlantic Ocean or border states or provinces that do.

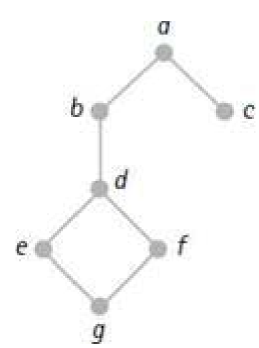
2. Find planar graphs with the smallest possible number of vertices that have chromatic numbers of 1, 2, 3, and 4.

3. What is the chromatic number of the graph representing the map of the United States? Explain your answer.

4. Draw a picture of the directed graph that corresponds to each of the following binary relations.

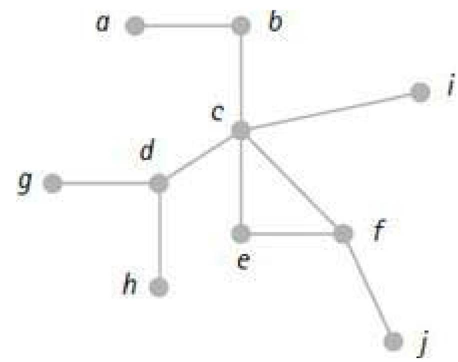
a. $\{(a, a), (b, b), (c, c)\}$.
b. $\{(a, b), (b, b), (b, c), (c, a)\}$.
c. The relation $\leq$ on the set $\{1, 2, 3\}$.

5. Given the following graph:



a. Write down all breadth-first traversals that start at vertex f.
b. Write down all depth-first traversals that start at vertex f.

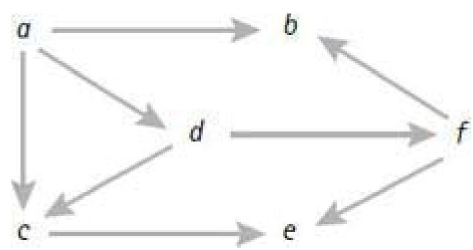
6. Given the following graph.



a. Write down one breadth-first traversal that starts at vertex f.

b. Write down one depth-first traversal that starts at vertex f.

7. Given the following graph:



a. Write down one breadth-first traversal that starts at vertex a.

b. Write down one depth-first traversal that starts at vertex a.

8. Convince yourself that in any set of six people there are three people who are either mutual friends or mutual strangers. *Hint:* Consider a complete graph with six vertices, where the vertices are people and an edge is colored red or blue to denote two friends or two strangers. Show that the graph must contain either a red-edged triangle or a blue-edged triangle.

Trees

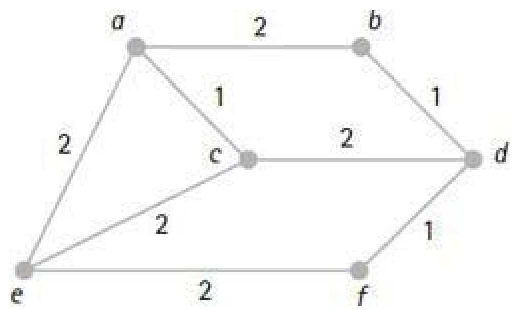
9. Given the algebraic expression $a \times (b + c) - (d / e)$. Draw a picture of the tree representation of this expression. Then convert the tree into a list representation of the expression.

10. Draw a picture of the ordered tree that is represented by the list

$$\langle a, \langle b, \langle c \rangle, \langle d, \langle e \rangle \rangle \rangle, \langle r, \langle s \rangle, \langle t \rangle \rangle, \langle x \rangle \rangle.$$

11. Draw a picture of a binary search tree containing the three-letter abbreviations for the 12 months of the year in dictionary order. Make sure that your tree has the least possible depth.

12. Find two distinct minimal spanning trees for the following weighted graph.



13. For the weighted graph in Example 1.44, find two distinct minimal spanning trees that are different from the spanning tree given.

14. For each case find a binary tree of height 2 with 5 nodes that satisfies the given condition with respect to traversals starting at the root.

a. Some depth-first traversal is the same as some breadth-first traversal.

b. Every depth-first traversal is different from every breadth-first traversal.

15. For each case find a binary tree of height 3 with 7 nodes that satisfies the given condition with respect to traversals starting at the root.

a. Some depth-first traversal is the same as some breadth-first traversal.

b. Every depth-first traversal is different from every breadth-first traversal.

16. Use examples to convince yourself of the following general facts.

a. The maximum number of leaves in a binary tree of height n is 2^n.

b. The maximum number of nodes in a binary tree of height n is $2^{n+1} - 1$.

17. A ternary tree is a tree with at most three children for each node. Use examples to convince yourself of the following general facts.

a. The maximum number of leaves in a ternary tree of height n is 3^n.

b. The maximum number of nodes in a ternary tree of height n is $(3^{n+1} -$

1)/2.

Path Problems

18. Try to find a necessary and sufficient condition for an undirected graph to have an Euler path. *Hint:* Look at the degrees of the vertices.

19. Try to find a necessary and sufficient condition for an undirected graph to have an Euler circuit. *Hint:* Look at the degrees of the vertices.

1.5 Chapter Summary

We normally prove things informally, and we use a variety of proof techniques: proof by exhaustive checking, conditional proof, proving the contrapositive, and proof by contradiction.

Sets are characterized by lack of order and no repeated occurrences of elements. There are easy techniques for comparing sets by subset and by equality. Sets can be combined by the operations of union, intersection, difference, and complement. Venn diagrams are useful for representing these operations. Two useful rules for counting sets are the union rule—also called the inclusion-exclusion principle—and the difference rule.

Bags—also called multisets—are characterized by lack of order, and they may contain repeated occurrences of elements.

Tuples are characterized by order, and they may contain repeated occurrences of elements. Many useful structures are related to tuples. Cartesian products of sets are collections of tuples. Lists are similar to tuples except that lists can be accessed only by head and tail. Strings are like lists except that elements from an alphabet are placed next to each other in juxtaposition. Languages are sets of strings and they can be combined by concatenating strings as well as by set operations. Relations are sets of tuples that are related in some way. A useful rule for counting tuples is the product rule.

Graphs are characterized by a set of vertices and a set of edges, where the edges may be undirected or directed. Graphs can be colored, and they can be traversed. Trees are special graphs that look like real trees. Prim's algorithm constructs a minimal spanning tree for an undirected, connected, weighted graph.

[1]Some people consider the natural numbers to be the set $\{1, 2, 3, \ldots\}$. If you are one of these people, then think of N as the nonnegative integers.

chapter 4

Equivalence, Order, and Inductive Proof

Good order is the foundation of all things.

—Edmund Burke (1729–1797)

Classifying things and ordering things are activities in which we all engage from time to time. Whenever we classify or order a set of things, we usually compare them in some way. That's how binary relations enter the picture.

In this chapter we'll discuss some special properties of binary relations that are useful for solving comparison problems. We'll introduce techniques to construct binary relations with the properties that we need. We'll discuss the idea of equivalence by considering properties of the equality relation. We'll also study the properties of binary relations that characterize our intuitive ideas about ordering. We'll also see that ordering is the fundamental ingredient needed to discuss inductive proof techniques.

chapter guide

4.1 (Properties of Binary Relations) introduces some of the desired properties of binary relations and shows how to construct new relations by composition and closure. We'll see how the results apply to solving path problems in graphs.

4.2 (Equivalence Relations) concentrates on the idea of equivalence. We'll see that equivalence is closely related to partitioning of sets. We'll show how to generate equivalence relations, we'll solve a typical equivalence problem, and we'll see an application for finding a spanning tree for a graph.

4.3 (Order Relations) introduces the idea of order. We'll discuss partial orders and how to sort them. We'll introduce well-founded orders and show some techniques for constructing them. Ordinal numbers are also introduced.

4.4 (Inductive Proof) introduces inductive proof techniques. We'll discuss the technique of mathematical induction for proving statements indexed by the natural numbers. Then we'll extend the discussion to inductive proof techniques for any well-founded set.

4.1 Properties of Binary Relations

Recall that the statement "R is a binary relation on the set A" means that R relates certain pairs of elements of A. Thus R can be represented as a set of ordered pairs (x, y), where $x, y \in A$. In other words, R is a subset of the Cartesian product $A \times A$. When $(x, y) \in R$, we also write $x\ R\ y$.

Binary relations that satisfy certain special properties can be very useful in solving computational problems. So let's discuss these properties.

Three Special Properties

For a binary relation R on a set A, we have the following definitions.

a. R is *reflexive* if $x\ R\ x$ for all $x \in A$.

b. R is *symmetric* if $x\ R\ y$ implies $y\ R\ x$ for all $x, y \in A$.

c. R is *transitive* if $x\ R\ y$ and $y\ R\ z$ implies $x\ R\ z$ for all $x, y, z \in A$.

Since a binary relation can be represented by a directed graph, we can describe the three properties in terms of edges: R is reflexive if there is an edge from x to x for each $x \in A$; R is symmetric if for each edge from x to y, there is also an edge from y to x. R is transitive if whenever there are edges from x to y and from y to z, there must also be an edge from x to z.

There are two more properties that we will also find to be useful.

Two More Properties

For a binary relation R on a set A, we have the following definitions.

a. R is *irreflexive* if $(x, x) \notin R$ for all $x \in A$.

b. R is *antisymmetric* if $x\ R\ y$ and $y\ R\ x$ implies $x = y$ for all $x, y \in A$.

From a graphical point of view we can say that R is irreflexive if there are no loop edges from x to x for all $x \in A$; and R is antisymmetric if whenever there is an edge from x to y with $x \neq y$, then there is no edge from y to x.

Many well-known relations satisfy one or more of the properties that we've been discussing. So we better look at a few examples.

example 4.1 Five Binary Relations

Here are some sample binary relations with the properties that they satisfy.

a. The equality relation on any set is reflexive, symmetric, transitive, and antisymmetric.

b. The $<$ relation on real numbers is transitive, irreflexive, and antisymmetric.

c. The $\leq$ relation on real numbers is reflexive, transitive, and antisymmetric.

d. The "is parent of" relation is irreflexive and antisymmetric.

e. The "has the same birthday as" relation is reflexive, symmetric, and transitive.

end example

4.1.1 Composition of Relations

Relations can often be defined in terms of other relations. For example, we can describe the "is grandparent of" relation in terms of the "is parent of" relation by saying that "a is grandparent of c" if and only if there is some b such that "a is parent of b" and "b is parent of c". This example demonstrates the fundamental idea of composing binary relations.

Definition of Composition

If R and S are binary relations, then the *composition* of R and S, which we denote by $R \circ S$, is the following relation:

$$R \circ S = \{(a, c) \mid (a, b) \in R \text{ and } (b, c) \in S \text{ for some element } b\}.$$

From a directed graph point of view, if we find an edge from a to b in the graph of R and we find an edge from b to c in the graph of S, then we must have an edge from a to c in the graph of $R \circ S$.

example 4.2 Grandparents

To construct the “isGrandparentOf” relation we can compose “isParentOf” with itself.

$$\text{isGrandparentOf} = \text{isParentOf} \circ \text{isParentOf}.$$

Similarly, we can construct the “isGreatGrandparentOf” relation by the following composition:

$$\text{isGreatGrandparentOf} = \text{isGrandparentOf} \circ \text{isParentOf}.$$

end example

example 4.3 Numeric Relations

Suppose we consider the relations “less,” “greater,” “equal,” and “notEqual” over the set $\mathbb{R}$ of real numbers. We want to compose some of these relations to see what we get. For example, let’s verify the following equality.

$$\text{greater} \circ \text{less} = \mathbb{R} \times \mathbb{R}.$$

For any pair (x, y), the definition of composition says that x (greater $\circ$ less) y if and only if there is some number z such that x greater z and z less y. We can write this statement more concisely as follows:

$$x\ (> \circ <)\ y \text{ iff there is some number } z \text{ such that } x > z \text{ and } z < y.$$

We know that for any two real numbers x and y there is always another number z that is less than both. So the composition must be the whole universe $\mathbb{R} \times \mathbb{R}$. Many combinations are possible. For example, it's easy to verify the following two equalities:

$$\begin{aligned} \text{equal} \circ \text{notEqual} &= \text{notEqual}, \\ \text{notEqual} \circ \text{notEqual} &= \mathbb{R} \times \mathbb{R}. \end{aligned}$$

end example

Other Combining Methods

Since relations are just sets (of ordered pairs), they can also be combined by the usual set operations of union, intersection, difference, and complement.

example 4.4 Combining Relations

The following samples show how we can combine some familiar numeric relations. Check out each one with a few example pairs of numbers.

$$\begin{aligned} \text{equal} \cap \text{less} &= \varnothing, \\ \text{equal} \cap \text{lessOrEqual} &= \text{equal}, \\ (\text{lessOrEqual})' &= \text{greater}, \\ \text{greaterOrEqual} - \text{equal} &= \text{greater}, \\ \text{equal} \cup \text{greater} &= \text{greaterOrEqual}, \\ \text{less} \cup \text{greater} &= \text{notEqual}. \end{aligned}$$

end example

Let's list some fundamental properties of combining relations. We'll leave the proofs of these properties as exercises.

Properties of Combining Relations **(4.1)**

a. $R \circ (S \circ T) = (R \circ S) \circ T$. (associativity)

b. $R \circ (S \cup T) = R \circ S \cup R \circ T$.

c. $R \circ (S \cap T) \subseteq R \circ S \cap R \circ T$.

Notice that Part (c) is stated as a set containment rather than an equality. For example, let R, S, and T be the following relations:

$$R = \{(a, b), (a, c)\}, S = \{(b, b)\}, T = \{(b, c), (c, b)\}.$$

Then $S \cap T = \varnothing$, $R \circ S = \{(a, b)\}$, and $R \circ T = \{(a, c), (a, b)\}$. Therefore

$$R \circ (S \circ T) = \varnothing \text{ and } R \circ S \cap R \circ T = \{(a, b)\}.$$

So (4.1c) isn't always an equality. But there are cases in which equality holds. For example, if $R = \varnothing$ or if $S \subseteq T$, then (4.1c) is an equality.

Representations

If R is a binary relation on A, then we'll denote the composition of R with itself n times by writing

$$R^n.$$

For example, if we compose isParentOf with itself, we get some familiar names as follows:

$$\begin{aligned} \text{isParentOf}^2 &= \text{isGrandparentOf}, \\ \text{isParentOf}^3 &= \text{isGreatGrandparentOf}. \end{aligned}$$

We mentioned in Chapter 1 that binary relations can be thought of as digraphs and, conversely, that digraphs can be thought of as binary relations. In other words, we can think of (x, y) as an edge from x to y in a digraph and as a member of a binary relation. So we can talk about the digraph of a binary relation.

An important and useful representation of R^n is as the digraph consisting of all edges (x, y) such that there is a path of length n from x to y. For example, if $(x, y) \in R^2$, then $(x, z), (z, y) \in R$ for some element z. This says that there is a path of length 2 from x to y in the digraph of R.

example 4.5 Compositions

Let $R = \{(a, b), (b, c), (c, d)\}$. The digraphs shown in Figure 4.1 are the digraphs for the three relations R, R^2, and R^3.

end example

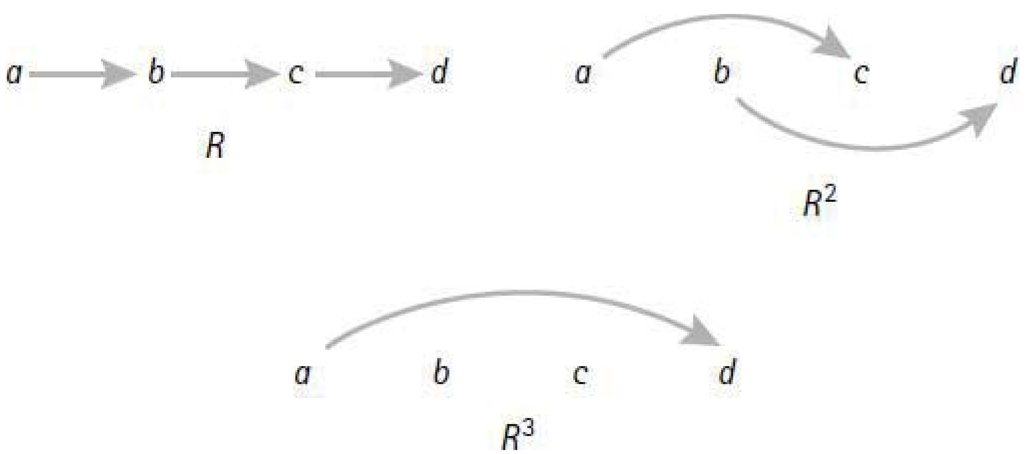


Figure 4.1 Composing a relation.

Let's give a more precise definition of R^n using induction. (Notice the interesting choice for R^0.)

$$R^0 = \{(a, a) \mid a \in A\} \quad \text{(basic equality)}$$
$$R^{n+1} = R^n \circ R.$$

We defined R^0 as the basic equality relation because we want to infer the equality $R^1 = R$ from the definition. To see this, observe the following evaluation of R^1:

$$R^1 = R^{0+1} = R^0 \circ R = \{(a, a) \mid a \in A\} \circ R = R.$$

We also could have defined $R^{n+1} = R \circ R^n$ instead of $R^{n+1} = R^n \circ R$ because composition of binary operations is associative by (4.1a).

Let's note a few other interesting relationships between R and R^n.

Inheritance Properties **(4.2)**

a. If R is reflexive, then R^n is reflexive.

b. If R is symmetric, then R^n is symmetric.

c. If R is transitive, then R^n is transitive.

On the other hand, if R is irreflexive, then it may not be the case that R^n is irreflexive. Similarly, if R is antisymmetric, it may not be the case that R^n is antisymmetric. We'll examine these statements in the exercises.

example 4.6 Integer Relations

Let $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x + y \text{ is odd}\}$. We'll calculate R^2 and R^3. To calculate R^2, we'll examine an arbitrary element $(x, y) \in R^2$. This means there is an element z such that $(x, z) \in R$ and $(z, y) \in R$. So $x + z$ is odd and $z + y$ is odd. We know that a sum is odd if and only if one number is even and the other number is odd. If x is even, then since $x + z$ is odd, it follows that z is odd. So, since $z + y$ is odd, it follows that y is even. Similarly, if x is odd, the same kind of reasoning shows that y is odd. So we have

$$R^2 = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x \text{ and } y \text{ are both even or both odd}\}.$$

To calculate R^3, we'll examine an arbitrary element $(x, y) \in R^3$. This means there is an element z such that $(x, z) \in R$ and $(z, y) \in R^2$. In other words, $x + z$ is odd and z and y are both even or both odd. If x is even, then since $x + z$ is odd, it follows that z is odd. So y must be odd. Similarly, if x is odd, the same kind of reasoning shows that y is even. So if $(x, y) \in R^3$, then one of x and y is even and the other is odd. In other words, $x + y$ is odd. Therefore

$$R^3 = R.$$

We don't have to go to higher powers now because, for example,

$$R^4 = R^3 \circ R = R \circ R = R^2.$$

end example

4.1.2 Closures

We've seen how to construct a new binary relation by composing two existing binary relations. Now we'll see how to construct a new binary relation by adding pairs of elements to an existing binary relation so that the new relation satisfies some particular property. For example, from the "isParentOf" relation for a family, we could construct the "isAncestorOf" relation for the family by adding the isGrandParentOf pairs, the isGreatGrandParentOf pairs, and so on. But for some properties this process cannot always be done. For example, if $R = \{(a, b), (b, a)\}$, then R is symmetric, so it cannot be a subset of any antisymmetric relation. To discuss this further we need to introduce the idea of closure.

Definition of Closure

If R is a binary relation and p is a property that can be satisfied by adding pairs of elements to R, then the *p closure* of R is the smallest binary relation containing R that has property p.

Three properties of interest for which closures always exist are reflexive, symmetric, and transitive. To introduce each of the three closures we'll use the following relation on the set $A = \{a, b, c\}$:

$$R = \{(a, a), (a, b), (b, a), (b, c)\}.$$

Notice that R is not reflexive, not symmetric, and not transitive. So the closures of R that we construct will all contain R as a proper subset.

Reflexive Closure

If R is a binary relation on A, then the *reflexive closure* of R, which we'll denote by $r(R)$, can be constructed by including all pairs (x, x) that are not already in R. Recall that the relation $\{(x, x) \mid x \in A\}$ is called the equality relation on A and it is also denoted by R^0. So we can say that

$$r(R) = R \cup R^0.$$

In our example, the pairs (b, b) and (c, c) are missing from R. So $r(R)$ is R together with these two pairs.

$$r(R) = \{(a, a), (a, b), (b, a), (b, c), (b, b), (c, c)\}.$$

Symmetric Closure

If R is a binary relation, then the *symmetric closure* of R, which we'll denote by $s(R)$, must include all pairs (x, y) for which $(y, x) \in R$. The set $\{(x, y) \mid (y, x) \in R\}$ is called the *converse* of R, which we'll denote by R^c. So we can say that

$$s(R) = R \cup R^c.$$

Notice that R is symmetric if and only if $R = R^c$.

In our example, the only problem is with the pair $(b, c) \in R$. Once we include the pair (c, b) we'll have $s(R)$.

$$s(R) = \{(a, a), (a, b), (b, a), (b, c), (c, b)\}.$$

Transitive Closure

If R is a binary relation, then the *transitive closure* of R is denoted by $t(R)$. We'll use our example to show how to construct $t(R)$. Notice that R contains the pairs (a, b) and (b, c), but (a, c) is not in R. Similarly, R contains the pairs (b, a) and (a, b), but (b, b) is not in R. So $t(R)$ must contain the pairs (a, c) and (b, b). Is there some relation that we can union with R that will add the two needed pairs? The answer is yes, it's R^2. Notice that

$$R^2 = \{(a, a), (a, b), (b, a), (b, b), (a, c)\}.$$

It contains the two missing pairs along with three other pairs that are already in R. Thus we have

$$t(R) = R \cup R^2 = \{(a, a), (a, b), (b, a), (b, c), (a, c), (b, b)\}.$$

To get some further insight into constructing the transitive closure, we need to look at another example. Let $A = \{a, b, c, d\}$, and let R be the following relation.

$$R = \{(a, b), (b, c), (c, d)\}.$$

To compute $t(R)$, we need to add the three pairs (a, c), (b, d), and (a, d). In this case, $R^2 = \{(a, c), (b, d)\}$. So the union of R with R^2 is missing (a, d). Can we find another relation to union with R and R^2 that will add this missing pair? Notice that $R^3 = \{(a, d)\}$. So for this example, $t(R)$ is the union

$$\begin{aligned} t(R) &= R \cup R^2 \cup R^3 \\ &= \{(a,b), (b,c), (c,d), (a,c), (b,d), (a,d)\}. \end{aligned}$$

As the examples show, $t(R)$ is a bit more difficult to construct than the other two closures.

Constructing the Three Closures

The three closures can be calculated by using composition and union. Here are the construction techniques.

Constructing Closures (4.3)

If R is a binary relation over a set A, then:

a. $r(R) = R \cup R^0$ (R^0 is the equality relation.)

b. $s(R) = R \cup R^c$ (R^c is the converse relation.)

c. $t(R) = R \cup R^2 \cup R^3 \cdots$.

d. If A is finite with n elements, then $t(R) = R \cup R^2 \cup \cdots \cup R^n$.

Let's discuss (4.3d). If $(x, y) \in R^m$, then there is a path of length m from x to y in the digraph representation of R. If $m > n$, then some element of A occurs at least twice on the path from x to y. So there is a shorter path from x to y. Thus $(x, y) \in R^k$ for some $k < m$. This argument can be repeated, if necessary, until we find that $(x, y) \in R^k$ for some $k \leq n$. So nothing new gets added to $t(R)$ by adding powers of R that are higher than n.

Sometimes we don't have to compute all the powers of R. For example, let $A = \{a, b, c, d, e\}$ and $R = \{(a, b), (b, c), (b, d), (d, e)\}$. The digraphs of R and $t(R)$ are drawn in Figure 4.2. Convince yourself that $t(R) = R \cup R^2 \cup R^3$. In other words, the relations R^4 and R^5 don't add anything new. In fact, you should verify that $R^4 = R^5 = \varnothing$.

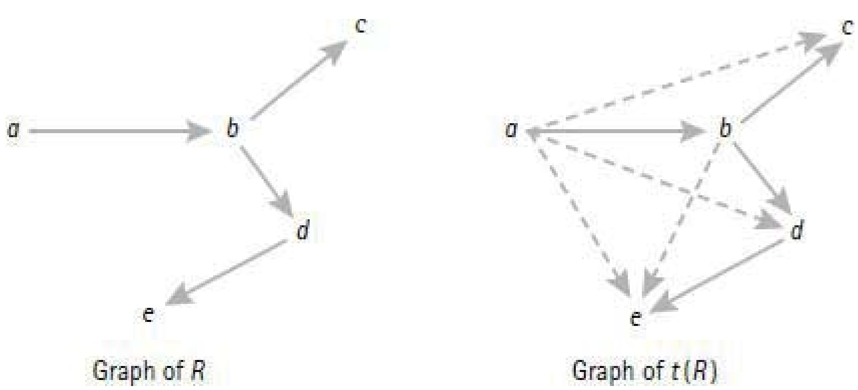


Figure 4.2 R and its transitive closure.

example 4.7 A Big Transitive Closure

Let $A = \{a, b, c\}$ and $R = \{(a, b), (b, c), (c, a)\}$. Then we have

$$R^2 = \{(a, c), (c, b), (b, a)\} \text{ and } R^3 = \{(a, a), (b, b), (c, c)\}.$$

So the transitive closure of R is the union

$$t(R) = R \cup R^2 \cup R^3 = A \times A.$$

end example

example 4.8 A Small Transitive Closure

Let $A = \{a, b, c\}$ and $R = \{(a, b), (b, c), (c, b)\}$. Then we have

$$R^2 = \{(a, c), (b, b), (c, c)\} \text{ and } R^3 = \{(a, b), (b, c), (c, b)\} = R.$$

So the transitive closure of R is the union of the sets, which gives

$$t(R) = \{(a, b), (b, c), (c, b), (a, c), (b, b), (c, c)\}.$$

end example

example 4.9 Generating Less-Than

Suppose $R = \{(x, x + 1) \mid x \in \mathbb{N}\}$. Then $R^2 = \{(x, x + 2) \mid x \in \mathbb{N}\}$. In general, for any natural number $k > 0$ we have

$$R^k = \{(x, x + k) \mid x \in \mathbb{N}\}.$$

Since $t(R)$ is the union of all these sets, it follows that $t(R)$ is the familiar "less" relation over $\mathbb{N}$. Just notice that if $x < y$, then $y = x + k$ for some k, so the pair (x, y) is in R^k.

end example

example 4.10 Closures of Numeric Relations

We'll list some closures for the numeric relations "less" and "notEqual" over the set $\mathbb{N}$ of natural numbers.

$$
\begin{aligned}
r\,(\text{less}) &= \text{lessOrEqual},\\
s\,(\text{less}) &= \text{notEqual},\\
t\,(\text{less}) &= \text{less},\\
r\,(\text{notEqual}) &= \mathbb{N} \times \mathbb{N},\\
s\,(\text{notEqual}) &= \text{notEqual},\\
t\,(\text{notEqual}) &= \mathbb{N} \times \mathbb{N}.
\end{aligned}
$$

end example

Properties of Closures

Some properties are retained by closures. For example, we have the following results, which we'll leave as exercises:

Inheritance Properties (4.4)

a. If R is reflexive, then $s(R)$ and $t(R)$ are reflexive.

b. If R is symmetric, then $r(R)$ and $t(R)$ are symmetric.

c. If R is transitive, then $r(R)$ is transitive.

Notice that (4.4c) doesn't include the statement "$s(R)$ is transitive" in its conclusion. To see why, we can let $R = \{(a, b), (b, c), (a, c)\}$. It follows that R is transitive. But $s(R)$ is not transitive because, for example, we have (a, b), $(b, a) \in s(R)$ and $(a, a) \notin s(R)$.

Sometimes, it's possible to take two closures of a relation and not worry about the order. Other times, we have to worry. For example, we might be interested in the double closure $r(s(R))$, which we'll denote by $rs(R)$. Do we get the same relation if we interchange r and s and compute $sr(R)$? The inheritance properties (4.4) should help us see that the answer is yes. Here are the facts:

Double Closure Properties (4.5)

a. $rt(R) = tr(R)$.

b. $rs(R) = sr(R)$.

c. $st(R) \subseteq ts(R)$.

Notice that (4.5c) is not an equality. To see why, let $A = \{a, b, c\}$, and consider the relation $R = \{(a, b), (b, c)\}$. Then $st(R)$ and $ts(R)$ are

$$st(R) = \{(a,b),(b,a),(b,c),(c,b),(a,c),(c,a)\}.$$
$$ts(R) = A \times A.$$

Therefore, $st(R)$ is a proper subset of $ts(R)$. Of course, there are also situations in which $st(R) = ts(R)$. For example, if $R = \{(a, a), (b, b), (c, c), (a, b), (b, c)\}$, then you can verify that $st(R) = ts(R)$.

Before we finish this discussion of closures, we should remark that the symbols R^+ and R^* are often used to denote the closures $t(R)$ and $rt(R)$.

4.1.3 Path Problems

Suppose we need to write a program that inputs two points in a city and outputs a bus route between the two points. A solution to the problem depends on the definition of "point." For example, if a point is any street intersection, then the solution may be harder than in the case in which a point is a bus stop.

This problem is an instance of a path problem. Let's consider some typical path problems in terms of a digraph.

Some Path Problems **(4.6)**

Given a digraph and two of its vertices i and j.

a. Find out whether there is a path from i to j. For example, find out whether there is a bus route from i to j.

b. Find a path from i to j. For example, find a bus route from i to j.

c. Find a path from i to j with the minimum number of edges. For example, find a bus route from i to j with the minimum number of stops.

d. Find a shortest path from i to j, where each edge has a nonnegative weight. For example, find the shortest bus route from i to j, where shortest might refer to distance or time.

e. Find the length of a shortest path from i to j. For example, find the number of stops (or the time or miles) on the shortest bus route from i to j.

Each problem listed in (4.6) can be phrased as a question and the same question is often asked over and over again (e.g., different people asking about the same bus route). So it makes sense to get the answers in advance if possible. We'll see how to solve each of the problems in (4.6).

Adjacency Matrix

A useful way to represent a binary relation R over a finite set A (equivalently, a digraph with vertices A and edges R) is as a special kind of matrix called an *adjacency matrix* (or incidence matrix). For ease of notation we'll assume that $A = \{1, \ldots, n\}$ for some n. The adjacency matrix for R is an n by n matrix M with entries defined as follows:

$$M_{ij} = \text{if } (i, j) \in R \text{ then } 1 \text{ else } 0.$$

example 4.11 An Adjacency Matrix

Consider the relation $R = \{(1, 2), (2, 3), (3, 4), (4, 3)\}$ over $A = \{1, 2, 3, 4\}$. We can represent R as a directed graph or as an adjacency matrix M. Figure 4.3 shows the two representations.

end example

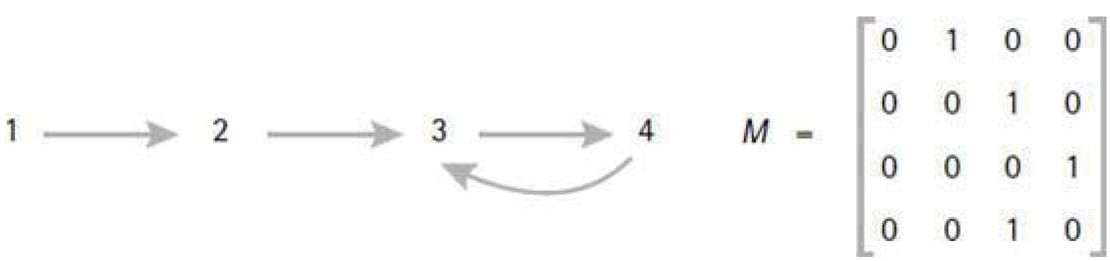


Figure 4.3 Directed graph and adjacency matrix.

If we look at the digraph in Figure 4.3, it's easy to see that R is neither reflexive, symmetric, nor transitive. We can see from the matrix M in Figure 4.3 that R is not reflexive because there is at least one zero on the main diagonal formed by the elements M_{ii}. Similarly, R is not symmetric because a reflection on the main diagonal is not the same as the original matrix. In other words, there are indices i and j such that $M_{ij} \neq M_{ji}$. R is not transitive, but there isn't any visual pattern in M that corresponds to transitivity.

It's an easy task to construct the adjacency matrix for $r(R)$: Just place 1's on the main diagonal of the adjacency matrix. It's also an easy task to construct the adjacency matrix for $s(R)$. We'll leave this one as an exercise.

Warshall's Algorithm for Transitive Closure

Let's look at an interesting algorithm to construct the adjacency matrix for $t(R)$. The idea, of course, is to repeat the following process until no new edges can be added to the adjacency matrix: If (i, k) and (k, j) are edges, then construct a new edge (i, j). The following algorithm to accomplish this feat with three **for**-loops is due to Warshall [1962].

Warshall's Algorithm for Transitive Closure **(4.7)**

Let M be the adjacency matrix for a relation R over $\{1, \ldots, n\}$. The algorithm replaces M with the adjacency matrix for $t(R)$.

```
for k := 1 to n do
    for i := 1 to n do
        for j := 1 to n do
            if (M_ik = M_kj = 1) then M_ij := 1
od od od
```

example 4.12 Applying Warshall's Algorithm

We'll apply Warshall's algorithm to find the transitive closure of the relation R given in Example 4.11. So the input to the algorithm will be the adjacency matrix M for R shown in Figure 4.3. The four matrices in Figure 4.4 show how Warshall's algorithm transforms M into the adjacency matrix for $t(R)$. Each matrix represents the value of M for the given value of k after the inner i and j loops have executed. To get some insight into how Warshall's algorithm works, draw the four digraphs for the adjacency matrices in Figure 4.4.

$$M \to \overset{k=1}{\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}} \to \overset{k=2}{\begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}} \to \overset{k=3}{\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}} \to \overset{k=4}{\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}}$$

Figure 4.4 Matrix transformations via Warshall's algorithm.

end example

Now we have an easy way find out whether there is a path from i to j in a digraph. Let R be the set of edges in the digraph. First we represent R as an adjacency matrix. Then we apply Warshall's algorithm to construct the adjacency matrix for $t(R)$. Now we can check to see whether there is a path from i to j in the original digraph by checking M_{ij} in the adjacency matrix M for $t(R)$. So we have all the solutions to problem (4.6a).

Floyd's Algorithm for Length of Shortest Path

Let's look at problem (4.6e). Can we compute the length of a shortest path in a weighted digraph? Sure. Let R denote the set of edges in the digraph. We'll represent the digraph as a *weighted adjacency matrix* M as follows: First of all, we set $M_{ii} = 0$ for $1 \le i \le n$ because we're not interested in the shortest path from i to itself. Next, for each edge $(i, j) \in R$ with $i \neq j$, we set M_{ij} to be the nonnegative weight for that edge. Lastly, if $(i, j) \notin R$ with $i \neq j$, then we set $M_{ij} = \infty$, where ∞ represents some number that is larger than the sum of all the weights on all the edges of the digraph.

example 4.13 A Weighted Adjacency Matrix

The diagram in Figure 4.5 represents the weighted adjacency matrix M for a weighted digraph over the vertex set $\{1, 2, 3, 4, 5, 6\}$.

end example

Now we can present an algorithm to compute the shortest distances between vertices in a weighted digraph. The algorithm, due to Floyd [1962], modifies the weighted adjacency matrix M so that M_{ij} is the shortest distance between distinct vertices i and j. For example, if there are two paths from i to j, then the entry M_{ij} denotes the smaller of the two path weights. So again, transitive closure comes into play. Here's the algorithm.

	1	2	3	4	5	6
1	0	10	10	∞	20	10
2	∞	0	∞	30	∞	∞
3	∞	∞	0	30	∞	∞
4	∞	∞	∞	0	∞	∞
5	∞	∞	∞	40	0	∞
6	∞	∞	∞	∞	5	0

Figure 4.5 Sample weighted adjacency matrix.

Floyd's Algorithm for Shortest Distances (4.8)

Let M be the weighted adjacency matrix for a weighted digraph over the set $\{1, \ldots, n\}$. The algorithm replaces M with a weighted adjacency matrix that represents the shortest distances between distinct vertices.

```
for k := 1 to n do
    for i := 1 to n do
        for j := 1 to n do
            M_ij := min{M_ij, M_ik + M_kj}
od od od
```

example **4.14 Applying Floyd's Algorithm**

We'll apply Floyd's algorithm to the weighted adjacency matrix in Figure 4.5. The result is given in Figure 4.6. The entries M_{ij} that are not zero and not ∞ represent the minimum distances (weights) required to travel from i to j in the original digraph.

end example

Let's summarize our results so far. Algorithm (4.8) creates a matrix M that allows us to easily answer two questions: Is there a path from i to j for distinct vertices i and j? Yes, if $M_{ij} \neq \infty$. What is the distance of a shortest path from i to j? It's M_{ij} if $M_{ij} \neq \infty$.

	1	2	3	4	5	6
1	0	10	10	40	15	10
2	∞	0	∞	30	∞	∞
3	∞	∞	0	30	∞	∞
4	∞	∞	∞	0	∞	∞
5	∞	∞	∞	40	0	∞
6	∞	∞	∞	45	5	0

Figure 4.6 The result of Floyd's algorithm.

Floyd's Algorithm for Finding the Shortest Path

Now let's try to find a shortest path. We can make a slight modification to (4.8) to compute a "path" matrix P, which will hold the key to finding a shortest path. We'll initialize P to be all zeros. The algorithm will modify P so that $P_{ij} = 0$ means that the shortest path from i to j is the edge from i to j and $P_{ij} = k$ means that a shortest path from i to j goes through k. The modified algorithm, which computes M and P, is stated as follows:

Shortest Distances and Shortest Paths Algorithm

(4.9)

Let M be the weighted adjacency matrix for a weighted digraph over the set $\{1, \ldots, n\}$. Let P be the n by n matrix of zeros. The algorithm replaces M by a matrix of shortest distances and it replaces P by a path matrix.

```
for k := 1 to n do
    for i := 1 to n do
        for j := 1 to n do
            if M_ik + M_kj < M_ij then
                M_ij := M_ik + M_kj ;
                P_ij := k
od od od fi
```

example 4.15 The Path Matrix

We'll apply (4.9) to the weighted adjacency matrix in Figure 4.5. The algorithm produces the matrix M in Figure 4.6, and it produces the path matrix P given in Figure 4.7.

For example, the shortest path between 1 and 4 passes through 2 because $P_{14} = 2$. Since $P_{12} = 0$ and $P_{24} = 0$, the shortest path between 1 and 4 consists of the sequence 1, 2, 4. Similarly, the shortest path between 1 and 5 is the sequence 1, 6, 5, and the shortest path between 6 and 4 is the sequence 6, 5, 4. So once we have matrix P from (4.9), it's an easy matter to compute a shortest path between two points. We'll leave this as an exercise.

	1	2	3	4	5	6
1	0	0	0	2	6	0
2	0	0	0	0	0	0
3	0	0	0	0	0	0
4	0	0	0	0	0	0
5	0	0	0	0	0	0
6	0	0	0	5	0	0

Figure 4.7 A path matrix.

end example

Let's make a few observations about Example 4.15. We should note that there is another shortest path from 1 to 4, namely, 1, 3, 4. The algorithm picked 2 as the intermediate point of the shortest path because the outer index k increments from 1 to n. When the computation got to $k = 3$, the value M_{14} had already been set to the minimal value, and P_{24} had been set to 2. So the condition of the if-then statement was false, and no changes were made. Therefore, P_{ij} gets the value of k closest to 1 whenever there are two or more values of k that give the same value to the expression $M_{ik} + M_{kj}$, and that value is less than M_{ij}.

Application Notes

Before we finish with this topic, let's make a couple of comments. If we have a digraph that is not weighted, then we can still find shortest distances and shortest paths with (4.8) and (4.9). Just let each edge have weight 1. Then the matrix M produced by either (4.8) or (4.9) will give us the length of a shortest path, and the matrix P produced by (4.9) will allow us to find a path of shortest length.

If we have a weighted graph that is not directed, then we can still use (4.8) and (4.9) to find shortest distances and shortest paths. Just modify the weighted adjacency matrix M as follows: For each edge between i and j having weight d, set $M_{ij} = M_{ji} = d$.

Exercises

Properties

1. Write down all of the properties that each of the following binary relations satisfies from among the five properties reflexive, symmetric, transitive, irreflexive, and antisymmetric.

a. The similarity relation on the set of triangles.

b. The congruence relation on the set of triangles.

c. The relation on people that relates people with the same parents.

d. The subset relation on sets.

e. The if and only if relation on the set of statements that may be true or false.

f. The relation on people that relates people with bachelor's degrees in computer science.

g. The "is brother of" relation on the set of people.

h. The "has a common national language with" relation on countries.

i. The "speaks the primary language of" relation on the set of people.

j. The "is father of" relation on the set of people.

2. Write down all of the properties that each of the following relations satisfies from among the properties reflexive, symmetric, transitive, irreflexive, and antisymmetric.

a. $R = \{(a, b) \mid a^2 + b^2 = 1\}$ over the real numbers.

b. $R = \{(a, b) \mid a^2 = b^2\}$ over the real numbers.

c. $R = \{(x, y) \mid x \bmod y = 0 \text{ and } x, y \in \{1, 2, 3, 4\}\}$.

d. $R = \{(x, y) \mid x \text{ divides } y\}$ over the positive integers.

e. $R = \{(x, y) \mid \gcd(x, y) = 1\}$ over the positive integers.

3. Explain why each of the following relations has the properties listed.

a. The empty relation $\varnothing$ over any set is irreflexive, symmetric, antisymmetric, and transitive.

b. For any set A, the universal relation $A \times A$ is reflexive, symmetric, and transitive. If $|A| = 1$, then $A \times A$ is also antisymmetric.

4. For each of the following conditions, find the smallest relation over the set $A = \{a, b, c\}$ that satisfies the stated properties.

a. Reflexive but not symmetric and not transitive.

b. Symmetric but not reflexive and not transitive.

c. Transitive but not reflexive and not symmetric.

d. Reflexive and symmetric but not transitive.

e. Reflexive and transitive but not symmetric.

f. Symmetric and transitive but not reflexive.

g. Reflexive, symmetric, and transitive.

Composition

5. Write down suitable names for each of the following compositions.

a. isChildOf $\circ$ isChildOf.

b. isSisterOf $\circ$ isParentOf.

c. isSonOf $\circ$ isSiblingOf.

d. isChildOf $\circ$ isSiblingOf $\circ$ isParentOf.

6. Suppose we define $x \; R \; y$ to mean "x is the father of y and y has a brother." Write R as the composition of two well-known relations.

7. For each of the following properties, find a binary relation R such that R has the property but R^2 does not.

a. Irreflexive.

b. Antisymmetric.

8. Given the relation "less" over the natural numbers $\mathbb{N}$, describe each of the following compositions as a set of the form $\{(x, y) \mid \text{property}\}$.

a. less $\circ$ less.

b. less $\circ$ less $\circ$ less.

9. Given the three relations "less," "greater," and "notEqual" over the natural numbers $\mathbb{N}$, find each of the following compositions.

a. less $\circ$ greater.

b. greater $\circ$ less.

c. notEqual $\circ$ less.
d. greater $\circ$ notEqual.

10. Let $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x + y \text{ is even}\}$. Find R^2.

Closure

11. Describe the reflexive closure of the empty relation $\varnothing$ over a set A.

12. Find the symmetric closure of each of the following relations over the set $\{a, b, c\}$.

a. $\varnothing$.
b. $\{(a, b), (b, a)\}$.
c. $\{(a, b), (b, c)\}$.
d. $\{(a, a), (a, b), (c, b), (c, a)\}$.

13. Find the transitive closure of each of the following relations over the set $\{a, b, c, d\}$.

a. $\varnothing$.
b. $\{(a, b), (a, c), (b, c)\}$.
c. $\{(a, b), (b, a)\}$.
d. $\{(a, b), (b, c), (c, d), (d, a)\}$.

14. Let $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x + y \text{ is odd}\}$. Use the results of Example 4.6 to calculate $t(R)$.

15. Find an appropriate name for the transitive closure of each of the following relations.

a. isParentOf.
b. isChildOf.
c. $\{(x + 1, x) \mid x \in \mathbb{N}\}$.

Path Problems

16. Suppose G is the following weighted digraph, where the triple (i, j, d) represents edge (i, j) with distance d:

$\{(1, 2, 20), (1, 4, 5), (2, 3, 10), (3, 4, 10), (4, 3, 5), (4, 2, 10)\}$.

a. Draw the weighted adjacency matrix for G.

b. Use (4.9) to compute the two matrices representing the shortest distances and the shortest paths in G.

17. Write an algorithm to compute the shortest path between two points of a weighted digraph from the matrix P produced by (4.9).

18. How many distinct path matrices can describe the shortest paths in the following graph, where it is assumed that all edges have weight = 1?

19. Write algorithms to perform each of the following actions for a binary relation R represented as an adjacency matrix.

a. Check R for reflexivity.

b. Check R for symmetry.

c. Check R for transitivity.

d. Compute $r(R)$.

e. Compute $s(R)$.

Proofs and Challenges

20. For each of the following properties, show that if R has the property, then so does R^2.

a. Reflexive.

b. Symmetric.

c. Transitive.

21. For the "less" relation over $\mathbb{N}$, show that $st(\text{less}) \neq ts(\text{less})$.

22. Prove each of the following statements about binary relations.

a. $R \circ (S \circ T) = (R \circ S) \circ T$. (associativity)

b. $R \circ (S \cup T) = R \circ S \cup R \circ T$.

c. $R \circ (S \cap T) \subseteq R \circ S \cap R \circ T$.

23. Let A be a set, R be any binary relation on A, and E be the equality relation on A. Show that $E \circ R = R \circ E = R$.

24. Prove each of the following statements about a binary relation R over a set A.

a. If R is reflexive, then $s(R)$ and $t(R)$ are reflexive.
b. If R is symmetric, then $r(R)$ and $t(R)$ are symmetric.
c. If R is transitive, then $r(R)$ is transitive.

25. Prove each of the following statements about a binary relation R over a set A.

a. $rt(R) = tr(R)$.
b. $rs(R) = sr(R)$.
c. $st(R) \subseteq ts(R)$.

26. A binary relation R over a set A is *asymmetric* if ($x\ R\ y$ and $y\ R\ x$) is false for all $x, y \in A$. Prove the following statements.

a. If R is asymmetric, then R is irreflexive.
b. If R is asymmetric, then R is antisymmetric.
c. R is asymmetric if and only if R is irreflexive and antisymmetric.

4.2 Equivalence Relations

The word "equivalent" is used in many ways. For example, we've all seen statements like "Two triangles are equivalent if their corresponding angles are equal." We want to find some general properties that describe the idea of "equivalence."

The Equality Problem

We'll start by discussing the idea of "equality" because, to most people, "equal" things are examples of "equivalent" things, whatever meaning is attached to the word "equivalent." Let's consider the following problem.

The Equality Problem

Write a computer program to check whether two objects are equal.

What is equality? Does it depend on the elements of the set? Why is equality important? What are some properties of equality? We all have an intuitive notion of what equality is because we use it all the time. Equality is important in computer science because programs use equality tests on data. If a programming language doesn't provide an equality test for certain data, then the programmer may need to implement such a test.

The simplest equality on a set A is basic equality: $\{(x, x) \mid x \in A\}$. But most of the time we use the word "equality" in a much broader context. For example, suppose A is the set of arithmetic expressions made from natural numbers and the symbol +. Thus A contains expressions like 3 + 7, 8, and 9 + 3 + 78. Most of us already have a pretty good idea of what equality means for these expressions. For example, we probably agree that 3 + 2 and 2 + 1 + 2 are equal. In other words, two expressions (*syntactic* objects) are equal if they have the same value (meaning or *semantics*), which is obtained by evaluating all + operations.

Are there some fundamental properties that hold for any definition of equality on a set A? Certainly we want to have $x = x$ for each element x in A (the basic equality on A). Also, whenever $x = y$, it ought to follow that $y = x$. Lastly, if $x = y$ and $y = z$, then $x = z$ should hold. Of course, these are the three properties reflexive, symmetric, and transitive.

Most equalities are more than just basic equality. That is, they equate different syntactic objects that have the same meaning. In these cases the symmetric and transitive properties are needed to convey our intuitive notion of equality. For example, the following statements are true if we let "=" mean "has the same value as":

If 2 + 3 = 1 + 4, then 1 + 4 = 2 + 3.
If 2 + 5 = 1 + 6 and 1 + 6 = 3 + 4, then 2 + 5 = 3 + 4.

4.2.1 Definition and Examples

Now we're ready to define equivalence. Any binary relation that is reflexive, symmetric, and transitive is called an *equivalence* relation. Sometimes people refer to an equivalence relation as an RST relation in order to remember the three properties.

Equivalence relations are all around us. Of course, the basic equality relation on any set is an equivalence relation. Similarly, the notion of equivalent triangles is an equivalence relation.

For another example, suppose we relate two books in the Library of Congress if their call numbers start with the same letter. (This is an instance in which it seems to be official policy to have a number start with a letter.) This relation is clearly an equivalence relation. Each book is related to itself (reflexive). If book A and book B have call numbers that begin with the same letter, then so do books B and A (symmetric). If books A and B have call numbers beginning with the same letter and books B and C have call numbers beginning with the same letter, then so do books A and C (transitive).

example 4.16 Sample Equivalence Relations

Here are a few more samples of equivalence relations, where the symbol ~ denotes each relation.

a. For the set of integers, let $x \sim y$ mean $x + y$ is even.

b. For the set of nonzero rational numbers, let $x \sim y$ mean $xy > 0$.

c. For the set of rational numbers, let $x \sim y$ mean $x - y$ is an integer.

d. For the set of triangles, let $x \sim y$ mean x and y are similar.

e. For the set of integers, let $x \sim y$ mean $x \bmod 4 = y \bmod 4$.

f. For the set of binary trees, let $x \sim y$ mean x and y have the same depth.

g. For the set of binary trees, let $x \sim y$ mean x and y have the same number of nodes.

h. For the set of real numbers, let $x \sim y$ mean $x^2 = y^2$.

i. For the set of people, let $x \sim y$ mean x and y have the same mother.

j. For the set of TV programs, let $x \sim y$ mean x and y start at the same time and day.

end example

Intersections and Equivalence Relations

We can always verify that a binary relation is an equivalence relation by checking that the relation is reflexive, symmetric, and transitive. But in some cases we can determine equivalence by other means. For example, we have the following intersection result, which we'll leave as an exercise.

Intersection Property of Equivalence (4.10)

If E and F are equivalence relations on the set A, then $E \cap F$ is an equivalence relation on A.

The practical use of (4.10) comes about when we notice that a relation $\sim$ on a set A is defined in the following form, where E and F are relations on A.

$$x \sim y \text{ iff } x\ E\ y \text{ and } x\ F\ y.$$

This is just another way of saying that $x \sim y$ iff $(x, y) \in E \cap F$. So if we can show that E and F are equivalence relations, then (4.10) tells us that $\sim$ is an equivalence relation.

example 4.17 Equivalent Binary Trees

Suppose we define the relation $\sim$ on the set of binary trees by

$x \sim y$ iff x and y have the same depth and the same number of nodes.

From Example 4.16 we know that "has the same depth as" and "has the same number of nodes as" are both equivalence relations. Therefore, $\sim$ is an equivalence relation.

end example

Equivalence Relations from Functions (Kernel Relations)

A very powerful technique for obtaining equivalence relations comes from the fact that any function defines a natural equivalence relation on its domain by relating elements that map to the same value. In other words, for any function $f : A \rightarrow B$, we obtain an equivalence relation $\sim$ on A by

$$x \sim y \text{ iff } f(x) = f(y).$$

It's easy to see that $\sim$ is an equivalence relation. The reflexive property follows because $f(x) = f(x)$ for all $x \in A$. The symmetric property follows because $f(x) = f(y)$ implies $f(y) = f(x)$. The transitive property follows because $f(x) = f(y)$ and $f(y) = f(z)$ implies $f(x) = f(z)$.

An equivalence relation defined in this way is called the *kernel relation* for f. Let's state the result for reference.

Kernel Relations **(4.11)**

If f is a function with domain A, then the relation $\sim$ defined by

$$x \sim y \text{ iff } f(x) = f(y)$$

is an equivalence relation on A, and it is called the *kernel relation* of f.

For example, notice that the relation given in Part (e) of Example 4.16 is the kernel relation for the function $f(x) = x \bmod 4$. Thus Part (e) of Example 4.16 is an equivalence relation by (4.11). Several other parts of Example 4.16 are also kernel relations. The nice thing about kernel relations is that they are always equivalence relations. So there is nothing to check. For example, we can use (4.11) to generalize Part (e) of Example 4.16 to the following important result.

Mod Function Equivalence **(4.12)**

If S is any set of integers and n is a positive integer, then the relation $\sim$ defined by

$$x \sim y \text{ iff } x \bmod n = y \bmod n$$

is an equivalence relation over S.

In many cases it's possible to show that a relation is an equivalence relation by rewriting its definition so that it is the kernel relation of some function.

example 4.18 A Numeric Equivalence Relation

Suppose we're given the relation $\sim$ defined on integers by

$$x \sim y \text{ if and only if } x - y \text{ is an even integer.}$$

We'll show that $\sim$ is an equivalence relation by writing it as the kernel relation of a function. Notice that $x - y$ is even if and only if x and y are both even or both odd. We can test whether an integer x is even or odd by checking whether $x \bmod 2 = 0$ or 1. So we can write our original definition of $\sim$ in terms of the mod function:

$$\begin{aligned} x \sim y \quad & \text{iff} \quad x - y \text{ is an even integer} \\ & \text{iff} \quad x \text{ and } y \text{ are both even or both odd} \\ & \text{iff} \quad x \bmod 2 = y \bmod 2. \end{aligned}$$

We can now conclude that $\sim$ is an equivalence relation because it's the kernel relation of the function f defined by $f(x) = x \bmod 2$.

end example

The Equivalence Problem

We can generalize the equality problem to the following more realistic problem of equivalence.

The Equivalence Problem

Write a computer program to check whether two objects are equivalent.

example 4.19 Binary Trees with the Same Structure

Suppose we need two binary trees to be equivalent whenever they have the same structure regardless of the values of the nodes. For binary trees S and T, let equiv(S, T) be true if S and T are equivalent and false otherwise. Here is a program to compute equiv.

$$\begin{aligned}\text{equiv}(S,T) = {} & \text{if } S = \langle\,\rangle \text{ and } T = \langle\,\rangle \text{ then True} \\ & \text{else if } S = \langle\,\rangle \text{ or } T = \langle\,\rangle \text{ then False} \\ & \text{else equiv}(\text{left}(S), \text{left}(T)) \text{ and equiv}(\text{right}(S), \text{right}(T)).\end{aligned}$$

end example

4.2.2 Equivalence Classes

The nice thing about an equivalence relation over a set is that it defines a natural way to group elements of the set into disjoint subsets. These subsets are called equivalence classes, and here's the definition.

Equivalence Class

Let R be an equivalence relation on a set S. If $a \in S$, then the *equivalence class* of a, denoted by $[a]$, is the subset of S consisting of all elements that are equivalent to a. In other words, we have

$$[a] = \{x \in S \mid x\ R\ a\}.$$

For example, we always have $a \in [a]$ because of the property $a\ R\ a$.

example 4.20 Equivalent Strings

Consider the relation $\sim$ defined on strings over the alphabet $\{a, b\}$ by

$$x \sim y \text{ iff } x \text{ and } y \text{ have the same length.}$$

Notice that $\sim$ is an equivalence relation because it is the kernel relation of the length function. Some sample equivalences are $abb \sim bab$ and $ba \sim aa$. Let's look at a few equivalence classes.

$$\begin{aligned}
[\Lambda] &= \{\Lambda\}, \\
[a] &= \{a, b\}, \\
[ab] &= \{ab, aa, ba, bb\}, \\
[aaa] &= \{aaa, aab, aba, baa, abb, bab, bba, bbb\}.
\end{aligned}$$

Notice that any member of an equivalence class can define the class. For example, we have

$$\begin{aligned}
[a] &= [b] = \{a, b\}, \\
[ab] &= [aa] = [ba] = [bb] = \{ab, aa, ba, bb\}.
\end{aligned}$$

end example

Equivalence classes enjoy a very nice property, namely that any two such classes are either equal or disjoint. Here is the result in more formal terms.

Property of Equivalences **(4.13)**

Let S be a set with an equivalence relation R. If $a, b \in S$, then either $[a] = [b]$ or $[a] \cap [b] = \varnothing$.

Proof: It suffices to show that $[a] \cap [b] \neq \varnothing$ implies $[a] = [b]$. If $[a] \cap [b] \neq \varnothing$, then there is a common element $c \in [a] \cap [b]$. It follows that cRa and cRb. From the symmetric and transitive properties of R, we conclude that aRb. To show that $[a] = [b]$, we'll show that $[a] \subseteq [b]$ and $[b] \subseteq [a]$. Let $x \in [a]$. Then xRa. Since aRb, the transitive property tells us that xRb, which implies that $x \in [b]$. Therefore, $[a] \subseteq [b]$. In an entirely similar manner we obtain $[b] \subseteq [a]$. Therefore, we have the desired result $[a] = [b]$. QED.

4.2.3 Partitions

By a *partition* of a set we mean a collection of nonempty subsets that are disjoint from each other and whose union is the whole set. For example, the set $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ can be partitioned in many ways, one of which consists of the following three subsets of S:

$$\{0, 1, 4, 9\}, \{2, 5, 8\}, \{3, 6, 7\}.$$

Notice that, if we wanted to, we could define an equivalence relation on S by saying that $x \sim y$ iff x and y are in the same set of the partition. In other words, we would have

$$\begin{aligned} [0] &= \{0, 1, 4, 9\}, \\ [2] &= \{2, 5, 8\}, \\ [3] &= \{3, 6, 7\}. \end{aligned}$$

We can do this for any partition of any set.

But something more interesting happens when we start with an equivalence relation on S. For example, let $\sim$ be the following relation on S:

$$x \sim y \text{ iff } x \bmod 4 = y \bmod 4.$$

This relation is an equivalence relation because it is the kernel relation of the function $f(x) = x \bmod 4$. Now let's look at some of the equivalence classes.

$$\begin{aligned} [0] &= \{0, 4, 8\}. \\ [1] &= \{1, 5, 9\}. \\ [2] &= \{2, 6\}. \\ [3] &= \{3, 7\}. \end{aligned}$$

Notice that these equivalence classes form a partition of S. This is no fluke. It always happens for any equivalence relation on any set S. To see this, notice that if $s \in S$, then $s \in [s]$, which says that S is the union of the equivalence classes. We also know from (4.13) that distinct equivalence classes are disjoint. Therefore, the set of equivalence classes forms a partition of S. Here's a summary of our discussion.

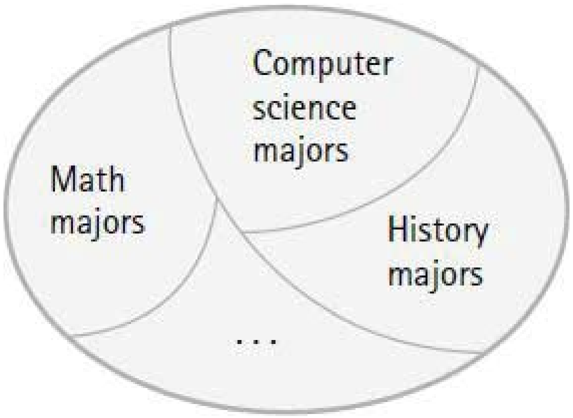


Figure 4.8 A partition of students.

Equivalence Relations and Partitions **(4.14)**

If R is an equivalence relation on the set S, then the equivalence classes form a partition of S. Conversely, if P is a partition of a set S, then there is an equivalence relation on S whose equivalence classes are sets of P.

For example, let S denote the set of all students at some university, and let M be the relation on S that relates two students if they have the same major. (Assume here that every student has exactly one major.) It's easy to see that M is an equivalence relation on S and each equivalence class is the set of all the students majoring in the same subject. For example, one equivalence class is the set of computer science majors. The partition of S is pictured by the Venn diagram in Figure 4.8.

example 4.21 Partitioning a Set of Strings

The relation from Example 4.20 is defined on the set $S = \{a, b\}^*$ of all strings over the alphabet $\{a, b\}$ by

$$x \sim y \text{ iff } x \text{ and } y \text{ have the same length.}$$

For each natural number n, the equivalence class $[a^n]$ contains all strings over $\{a, b\}$ that have length n. So we can write

$$S = \{a, b\}^* = [\Lambda] \cup [a] \cup [aa] \cup \ldots \cup [a^n] \cup \ldots.$$

end example

example 4.22 A Partition of the Natural Numbers

Let $\sim$ be the relation on the natural numbers defined by

$$x \sim y \text{ iff } \lfloor x/10 \rfloor = \lfloor y/10 \rfloor.$$

This is an equivalence relation because it is the kernel relation of the function $f(x) = \lfloor x/10 \rfloor$. After checking a few values we see that each equivalence class is a decade of numbers. For example,

$$[0] = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\},$$
$$[10] = \{10, 11, 12, 13, 14, 15, 16, 17, 18, 19\},$$

and in general, for any natural number n,

$$[10n] = \{10n, 10n + 1, \ldots, 10n + 9].$$

So we have $\mathbb{N} = [0] \cup [10] \cup \ldots \cup [10n] \cup \ldots.$

end example

example 4.23 Partitioning with Mod 5

Let R be the equivalence relation on the integers $\mathbb{Z}$ defined by

$$a \, R \, b \text{ iff } a \bmod 5 = b \bmod 5.$$

After some checking we see that the partition of $\mathbb{Z}$ consists of the following five equivalence classes

$$
\begin{aligned}
[0] &= \{\ldots -10, -5, 0, 5, 10, \ldots\}, \\
[1] &= \{\ldots -9, -4, 1, 6, 11, \ldots\}, \\
[2] &= \{\ldots -8, -3, 2, 7, 12, \ldots\}, \\
[3] &= \{\ldots -7, -2, 3, 8, 13, \ldots\}, \\
[4] &= \{\ldots -6, -1, 4, 9, 14, \ldots\}.
\end{aligned}
$$

Remember, it doesn't matter which element of a class is used to represent it. For example, $[0] = [5] = [-15]$. It is clear that the five classes are disjoint from each other and that $\mathbb{Z}$ is the union of the five classes.

end example

example 4.24 Software Testing

If the input data set for a program is infinite, then the program can't be tested on every input. However, every program has a finite number of instructions. So we should be able to find a finite data set to cause all instructions of the program to be executed. For example, suppose p is the following program, where x is an integer and q, r, and s represent other parts of the program:

$$
\begin{array}{ll}
p(x): & \textbf{if } x > 0 \textbf{ then } q(x) \\
& \textbf{else if } x \text{ is even } \textbf{then } r(x) \\
& \qquad \textbf{else } s(x) \\
& \qquad \textbf{fi} \\
& \textbf{fi}
\end{array}
$$

The condition "$x > 0$" causes a natural partition of the integers into the positives and the nonpositives. The condition "x is even" causes a natural partition of the nonpositives into the even nonpositives and the odd nonpositives. So we have a partition of the integers into the following three subsets:

$$\{1, 2, 3, \ldots\}, \{0, -2, -4, \ldots\}, \{-1, -3, -5, \ldots\}.$$

Now we can test the instructions in q, r, and s by picking three numbers, one from each set of the partition. For example, $p(1)$, $p(0)$, and $p(-1)$ will do the job. Of course, further partitioning may be necessary if q, r, or s contains further conditional statements. The equivalence relation induced by the partition relates two integers x and y if and only if $p(x)$ and $p(y)$ execute the same set of instructions.

end example

Refinement of a Partition

Suppose that P and Q are two partitions of a set S. If each set of P is a subset of a set in Q, then P is a *refinement* of Q. We also say P is *finer* than Q or Q is *coarser* than P. The finest of all partitions on S is the collection of singleton sets. The coarsest of all partitions of S is $\{S\}$.

For example, here is a listing of four partitions of $\{a, b, c, d\}$ that are successive refinements from the coarsest to finest:

$$\begin{array}{ll} \{\{a, b, c, d\}\} & \text{(coarsest)} \\ \{\{a, b\}, \{c, d\}\} & \\ \{\{a, b\}, \{c\}, \{d\}\} & \\ \{\{a\}, \{b\}, \{c\}, \{d\}\} & \text{(finest).} \end{array}$$

example 4.25 Partitioning with Mod 2

Let R be the relation over $\mathbb{N}$ defined by

$$a\ R\ b \text{ iff } a \bmod 2 = b \bmod 2.$$

Then R is an equivalence relation because it is the kernel relation of the function f defined by $f(x) = x \bmod 2$. The corresponding partition of $\mathbb{N}$ consists of the two subsets

$$[0] = \{0, 2, 4, 6, \ldots\},$$
$$[1] = \{1, 3, 5, 7, \ldots\}.$$

Can we find a refinement of this partition? Sure. Let T be defined by

$$a\ T\ b \text{ iff } a \bmod 4 = b \bmod 4.$$

T induces the following partition of $\mathbb{N}$ that is a refinement of the partition induced by R because we get the following four equivalence classes:

$$[0] = \{0, 4, 8, 12, \ldots\},$$
$$[1] = \{1, 5, 9, 13, \ldots\},$$
$$[2] = \{2, 6, 10, 14, \ldots\},$$
$$[3] = \{3, 7, 11, 15, \ldots\}.$$

This partition is indeed a refinement of the preceding partition. Can we find a refinement of this partition? Yes, because we can continue the process forever. Just let k be a power of 2 and define T_k by

$$a\ T_k\ b \text{ iff } a \bmod k = b \bmod k.$$

So the partition for each T_{2k} is a refinement of the partition for T_k.

end example

An Intersection Property

We noted in (4.10) that the intersection of equivalence relations over a set A is also an equivalence relation over A. It also turns out that the equivalence classes for the intersection are intersections of equivalence classes for the given relations. Here is the statement and we'll leave the proof as an exercise.

Intersection Property of Equivalence **(4.15)**

Let E and F be equivalence relations on a set A. Then the equivalence classes for the relation $E \cap F$ are of the form $[x] = [x]_E \cap [x]_F$, where $[x]_E$ and $[x]_F$ denote the equivalence classes of x for E and F, respectively.

example 4.26 Intersecting Equivalence Relations

Let $\sim$ be the relation on the natural numbers defined by

$$x \sim y \text{ iff } \lfloor x/10 \rfloor = \lfloor y/10 \rfloor \text{ and } x + y \text{ is even.}$$

Notice that $\sim$ is the intersection of two relations E and F, where $x\ E\ y$ means $\lfloor x/10 \rfloor = \lfloor y/10 \rfloor$ and $x\ F\ y$ means $x + y$ is even. We can observe that $x + y$ is even if and only if $x \bmod 2 = y \bmod 2$. So both E and F are kernel relations of functions and thus are equivalence relations. Therefore, $\sim$ is an equivalence relation by (4.10). We computed the equivalence classes for E and F in Examples 4.22 and 4.25. The equivalence classes for E are of the following form for each natural number n.

$$[10n] = \{10n, 10n + 1, \ldots, 10n + 9].$$

The equivalence classes for F are

$$[0] = \{0, 2, 4, 6, \ldots\},$$
$$[1] = \{1, 3, 5, 7, \ldots\}.$$

By (4.15) the equivalence classes for $\sim$ have the following form for each n:

$$[10n] \cap [0] = \{10n, 10n + 2, 10n + 4, 10n + 6, 10n + 8\},$$
$$[10n] \cap [1] = \{10n + 1, 10n + 3, 10n + 5, 10n + 7, 10n + 9\}.$$

end example

example **4.27 Solving the Equality Problem**

If we want to define an equality relation on a set S of objects that do not have any established meaning, then we can use the basic equality relation $\{(x, x) \mid x \in S\}$. On the other hand, suppose a meaning has been assigned to each element of S. We can represent the meaning by a mapping m from S to a set of values V. In other words, we have a function $m : S \to V$. It's natural to define two elements of S to be equal if they have the same meaning. That is, we define $x = y$ if and only if $m(x) = m(y)$. This equality relation is just the kernel relation of m.

For example, let S denote the set of arithmetic expressions made from nonempty unary strings and the symbol +. For example, some typical expressions in S are 1, 11, 111, 1+1, 11+111+1. Now let's assign a meaning to each expression in S. Let $m(1^n) = n$ for each positive natural number n. If $e + e'$ is an expression of S, we define $m(e + e') = m(e) + m(e')$. We'll assume that + is applied left to right. For example, the value of the expression 1 + 111 + 11 can be calculated as follows:

$$\begin{aligned} m(1+111+11) &= m((1+111)+11) \\ &= m(1+111)+m(11) \\ &= m(1)+m(111)+2 \\ &= 1+3+2 \\ &= 6. \end{aligned}$$

If we define two expressions of S to be equal when they have the same meaning, then the desired equality relation on S is the kernel relation of m. So the partition of S induced by the kernel relation of m consists of the sets of expressions with equal values. For example, the equivalence class [1111] contains the eight expressions

1+1+1+1, 1+1+11, 1+11+1, 11+1+1, 11+11, 1+111, 111+1, 1111.

end example

4.2.4 Generating Equivalence Relations

Any binary relation can be considered as the *generator* of an equivalence relation obtained by adding just enough pairs to make the result reflexive, symmetric, and transitive. In other words, we can take the reflexive, symmetric, and transitive closures of the binary relation.

Does the order that we take closures make a difference? For example, what about $str(R)$? An example will suffice to show that $str(R)$ need not be an equivalence relation. Let $A = \{a, b, c\}$ and $R = \{(a, b), (a, c), (b, b)\}$. Then

$$str(R) = \{(a, a), (b, b), (c, c), (a, b), (b, a), (a, c), (c, a)\}.$$

This relation is reflexive and symmetric, but it's not transitive. On the other hand, we have $tsr(R) = A \times A$, which is an equivalence relation. As the next result shows, $tsr(R)$ is always an equivalence relation.

The Smallest Equivalence Relation (4.16)

If R is a binary relation on A, then $tsr(R)$ is the smallest equivalence relation that contains R.

Proof: The inheritance properties of (4.4) tell us that $tsr(R)$ is an equivalence relation. To see that it's the smallest equivalence relation containing R, we'll let T be an arbitrary equivalence relation containing R. Since $R \subseteq T$ and T is reflexive, it follows that $r(R) \subseteq T$. Since $r(R) \subseteq T$ and T is symmetric, it follows that $sr(R) \subseteq T$. Since $sr(R) \subseteq T$ and T is transitive, it follows that $tsr(R) \subseteq T$. So $tsr(R)$ is contained in every equivalence relation that contains R. Thus it's the smallest equivalence relation containing R. QED.

example 4.28 Family Trees

Let R be the "is parent of" relation for a set of people. In other words, $(x, y) \in R$ iff x is a parent of y. Suppose we want to answer questions like the following:

Is x a descendant of y?
Is x an ancestor of y?

Are x and y related in some way?

What is the relationship between x and y?

Each of these questions can be answered from the given information by finding whether an appropriate path exists between x and y. But if we construct $t(R)$, then things get better because $(x, y) \in t(R)$ iff x is an ancestor of y. So we can find out whether x is an ancestor of y or x is a descendant of y by looking to see whether $(x, y) \in t(R)$ or $(y, x) \in t(R)$.

If we want to know whether x and y are related in some way, then we would have to look for paths in $t(R)$ taking each of x and y to a common ancestor. But if we construct $ts(R)$, then things get better because $(x, y) \in ts(R)$ iff x and y have a common ancestor. So we can find out whether x and y are related in some way by looking to see whether $(x, y) \in ts(R)$.

If x and y are related, then we might want to know the relationship. This question is asking for paths from x and y to a common ancestor, which can be done by searching $t(R)$ for the common ancestor and keeping track of each person along the way.

Notice also that the set of people can be partitioned into family trees by the equivalence relation $tsr(R)$. So the simple "is parent of" relation is the generator of an equivalence relation that constructs family trees.

end example

An Equivalence Problem

Suppose we have an equivalence relation over a set S that is generated by a given set of pairs. For example, the equivalence relation might be the family relationship "is related to" and the generators might be a set of parent-child pairs.

Can we represent the generators in such a way that we can find out whether two arbitrary elements of S are equivalent? If two elements are equivalent, can we find a sequence of generators to confirm the fact? The answer to both questions is yes. We'll present a solution due to Galler and Fischer [1964], which uses a special kind of tree structure to represent the equivalence classes.

The idea is to use the generating pairs to build the partition of S induced by the equivalence relation. For example, let $S = \{1, 2, \ldots, 10\}$, let $\sim$ denote the equivalence relation on S, and let the generators be the following pairs:

$$1 \sim 8,\ 4 \sim 5,\ 9 \sim 2,\ 4 \sim 10,\ 3 \sim 7,\ 6 \sim 3,\ 4 \sim 9.$$

To have something concrete in mind, let the numbers 1, 2, . . . , 10 be people, let $\sim$ be "is related to," and let the generators be "parent $\sim$ child" pairs.

The construction process starts by building the following ten singleton equivalence classes to represent the partition of S caused by the reflexive property $x \sim x$.

$$\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}, \{7\}, \{8\}, \{9\}, \{10\}.$$

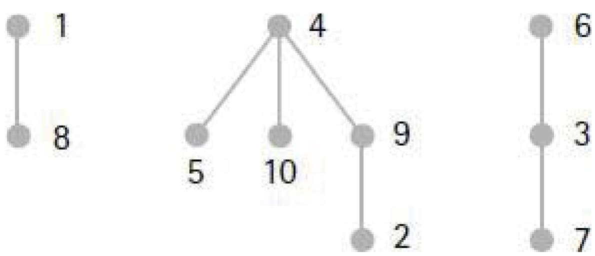


Figure 4.9 Equivalence classes as trees.

i	1	2	3	4	5	6	7	8	9	10
$p[i]$	0	9	6	0	4	0	3	1	4	4

Figure 4.10 Equivalence classes as an array.

Now we process the generators, one at a time. The generator $1 \sim 8$ is processed by forming the union of the equivalence classes that contain 1 and 8. In other words, the partition becomes

$$\{1, 8\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}, \{7\}, \{9\}, \{10\}.$$

Continuing in this manner to process the other generators, we eventually obtain the partition of S consisting of the following three equivalence classes.

$$\{1, 8\}, \{2, 4, 5, 9, 10\}, \{3, 6, 7\}.$$

Representing Equivalence Classes

To answer questions about an equivalence relation, we need to consider its representation. We can represent each equivalence class in the partition as a tree, where the generator $a \sim b$ will be processed by creating the branch "a is the parent of b." For our example, if we process the generators in the order in which they are written, then we obtain the three trees in Figure 4.9.

A simple way to represent these trees is with a 10-tuple (a 1-dimensional array of size 10) named p, where $p[i]$ denotes the parent of i. We'll let $p[i] = 0$ mean that i is a root. Figure 4.10 shows the three equivalence classes represented by p.

Now it's easy to answer the question "Is $a \sim b$?" Just find the roots of the trees to which a and b belong. If the roots are the same, the answer is yes. If the answer is yes, then there is another question, "Can you find a sequence of equivalences to show that $a \sim b$?" One way to do this is to locate one of the numbers, say b, and rearrange the tree to which b belongs so that b becomes the root. This can be done easily by reversing the links from b to the root. Once we have b at the root, it's an easy matter to read off the equivalences from a to b. We'll leave it as an exercise to construct an algorithm to do the reversing.

For example, if we ask whether $5 \sim 2$, we find that 5 and 2 belong to the same tree. So the answer is yes. To find a set of equivalences to prove that $5 \sim 2$, we can, for example, reverse the links from 2 to the root of the tree. The before and after pictures are given in Figure 4.11.

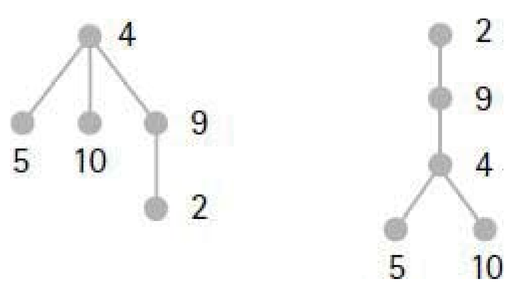


Figure 4.11 Proof that $5 \sim 2$.

Now it's an easy computation to traverse the tree from 5 to the root 2 and read off the equivalences $5 \sim 4$, $4 \sim 9$, and $9 \sim 2$.

Kruskal's Algorithm for Minimal Spanning Trees

In Chapter 1 we discussed Prim's algorithm to find a minimal spanning tree for a connected weighted undirected graph. Let's look an another such algorithm, due to Kruskal [1956], which uses equivalence classes.

The algorithm constructs a minimal spanning tree as follows: Starting with an empty tree, an edge $\{a, b\}$ of smallest weight is chosen from the graph. If there is no path in the tree from a to b, then the edge $\{a, b\}$ is added to the tree. This process is repeated with the remaining edges of the graph until the tree contains all vertices of the graph.

At any point in the algorithm, the edges in the spanning tree define an equivalence relation on the set of vertices of the graph. Two vertices a and b are equivalent iff there is a path between a and b in the tree. Whenever an edge $\{a, b\}$ is added to the spanning tree, the equivalence relation is modified by creating the equivalence class $[a] \cup [b]$. The algorithm ends when there is exactly one equivalence class consisting of all the vertices of the graph. Here are the steps of the algorithm.

Kruskal's Algorithm

1. Sort the edges of the graph by weight, and let L be the sorted list.

2. Let T be the minimal spanning tree and initialize $T := \varnothing$.

3. For each vertex v of the graph, create the equivalence class $[v] = \{v\}$.

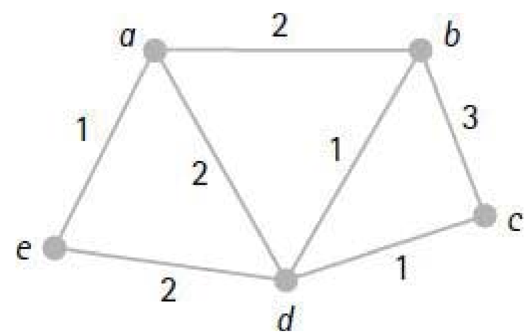
4. **while** there are 2 or more equivalence classes **do**

```
        L := tail(L);
        if [a] ≠ [b] then
            T := T ∪ {{a, b}};
            Replace the equivalence classes [a] and [b] by [a] ∪ [b]
        fi
    od
```

To implement the algorithm, we must find a representation for the equivalence classes. For example, we might use a parent array like the one we've been discussing.

example 4.29 Minimal Spanning Trees

We'll use Kruskal's algorithm to construct a minimal spanning tree for the following weighted graph:



To see how the algorithm works, we'll do a trace of each step. We'll assume that the edges have been sorted by weight in the following order:

$$\{a, e\}, \{b, d\}, \{c, d\}, \{a, b\}, \{a, d\}, \{e, d\}, \{b, c\}.$$

The following table shows the value of the spanning tree T and the equivalence classes at each step, starting with the initialization values.

Spanning Tree T	*Equivalence Classes*
$\{\}$	$\{a\}, \{b\}, \{c\}, \{d\}, \{e\}$
$\{\{a, e\}\}$	$\{a, e\}, \{b\}, \{c\}, \{d\}$
$\{\{a, e\}, \{b, d\}\}$	$\{a, e\}, \{b, d\}, \{c\}$
$\{\{a, e\}, \{b, d\}, \{c, d\}\}$	$\{a, e\}, \{b, c, d\}$
$\{\{a, e\}, \{b, d\}, \{c, d\}, \{a, b\}\}$	$\{a, b, c, d, e\}$

The algorithm stops because there is only one equivalence class. So T is a spanning tree for the graph.

end example

Exercises

Properties

1. Verify that each of the following relations is an equivalence relation.

a. $x \sim y$ iff x and y are points in a plane equidistant from a fixed point.
b. $s \sim t$ iff s and t are strings with the same occurrences of each letter.
c. $x \sim y$ iff $x + y$ is even, over the set of natural numbers.
d. $x \sim y$ iff $x - y$ is an integer, over the set of rational numbers.
e. $x \sim y$ iff $xy > 0$, over the set of nonzero rational numbers.

2. Each of the following relations is not an equivalence relation. In each case, find the properties that are not satisfied.

a. $a\ R\ b$ iff $a + b$ is odd, over the set of integers.
b. $a\ R\ b$ iff a/b is an integer, over the set of nonzero rational numbers.
c. $a\ R\ b$ iff $|a - b| \leq 5$, over the set of natural numbers.
d. $a\ R\ b$ iff either a mod 4 = b mod 4 *or* a mod 6 = b mod 6, over $\mathbb{N}$.
e. $a\ R\ b$ iff $x < a/10 < x + 1$ and $x \leq b/10 < x + 1$ for some integer x.

Equivalence Classes

3. For each of the following functions f with domain $\mathbb{N}$, describe the equivalence classes of the kernel relation of f.

a. $f(x) = 7$.
b. $f(x) = x$.
c. $f(x) = \text{floor}(x/2)$.
d. $f(x) = \text{floor}(x/3)$.
e. $f(x) = \text{floor}(x/4)$.
f. $f(x) = \text{floor}(x/k)$ for a fixed positive integer k.
g. $f(x) = \text{if } 0 \leq x \leq 10 \text{ then } 10 \text{ else } x - 1$.

4. For each of the following functions f, describe the equivalence classes of the kernel relation of f that partition the domain of f.

a. $f : \mathbb{Z} \rightarrow \mathbb{N}$ is defined by $f(x) = |x|$.
b. $f : \mathbb{R} \rightarrow \mathbb{Z}$ is defined by $f(x) = \text{floor}(x)$.

5. Describe the equivalence classes for each of the following relations on $\mathbb{N}$.

a. $x \sim y$ iff x mod $2 = y$ mod 2 *and* x mod $3 = y$ mod 3.

b. $x \sim y$ iff x mod $2 = y$ mod 2 *and* x mod $4 = y$ mod 4.

c. $x \sim y$ iff x mod $4 = y$ mod 4 *and* x mod $6 = y$ mod 6.

6. Given the following set of words.

$$\{rot, tot, root, toot, roto, toto, too, to, otto\}.$$

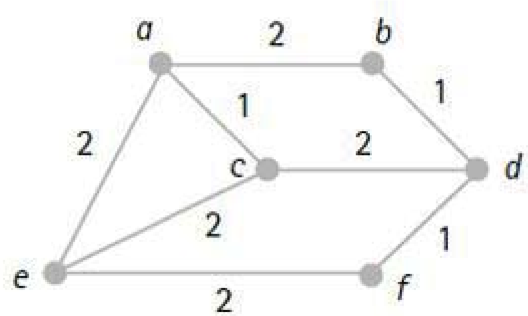
a. Let f be the function that maps a word to its set of letters. For the kernel relation of f, describe the equivalence classes.

b. Let f be the function that maps a word to its bag of letters. For the kernel relation of f, describe the equivalence classes.

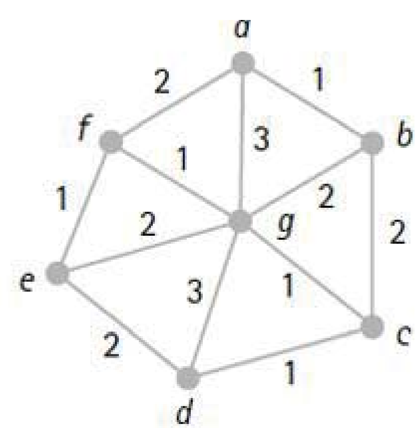
Spanning Trees

7. Use Kruskal's algorithm to find a minimal spanning tree for each of the following weighted graphs.

a.



b.



Proofs and Challenges

8. Let R be a relation on a set S such that R is symmetric and transitive and for each $x \in S$ there is an element $y \in S$ such that $x \ R \ y$. Prove that R is an equivalence relation (i.e., prove that R is reflexive).

9. Let E and F be equivalence relations on the set A, Show that $E \cap F$ is an equivalence relation on A.

10. Let E and F be equivalence relations on a set A and for each $x \in A$ let $[x]_E$ and $[x]_F$ denote the equivalence classes of x for E and F, respectively. Show that the equivalence classes for the relation $E \cap F$ are of the form $[x] = [x]_E \cap [x]_F$ for all $x \in A$.

11. Which relations among the following list are equal to $tsr(R)$, the smallest equivalence relation generated by R?

$$trs(R),\ str(R),\ srt(R),\ rst(R),\ rts(R).$$

12. In the equivalence problem we represented equivalence classes as a set of trees, where the nodes of the trees are the numbers $1, 2, \ldots, n$. Suppose the trees are represented by an array $p[1], \ldots, p[n]$, where $p[i]$ is the parent of i. Suppose also that $p[i] = 0$ when i is a root. Write a procedure that takes a node i and rearranges the tree that i belongs to so that i is the root, by reversing the links from the root to i.

13. (*Factoring a Function*). An interesting consequence of equivalence relations and partitions is that any function f can be factored into a composition of two functions, one an injection and one a surjection. For a function $f : A \to B$, let P be the partition of A by the kernel relation of f. Then define the function $s : A \to P$ by $s(a) = [a]$ and define $i : P \to B$ by $i([a]) = f(a)$. Prove that s is a surjection, i is an injection, and $f = i \circ s$.

4.3 Order Relations

Each day we see the idea of "order" used in many different ways. For example, we might encounter the expression $1 < 2$. We might notice that someone is older than someone else. We might be interested in the third component of the tuple (x, d, c, m). We might try to follow a recipe. Or we might see that the word "aardvark" resides at a certain place in the dictionary. The concept of order occurs in many different forms, but they all have the common idea of some object preceding another object.

Two Essential Properties of Order

Let's try to formally describe the concept of order. To have an ordering, we need a set of elements together with a binary relation having certain properties. What are these properties?

Well, our intuition tells us that if a, b, and c are objects that are ordered so that a precedes b and b precedes c, then we certainly want a to precede c. In other words, an ordering should be transitive. For example, if a, b, and c are natural numbers and $a < b$ and $b < c$, then we have $a < c$.

Our intuition also tells us that we don't want distinct objects preceding each other. In other words, if a and b are distinct objects and a precedes b, then b can't precede a. In still other words, if a precedes b and b precedes a then we better have $a = b$. For example, if a, b, and c are natural numbers and $a \leq b$ and $b \leq a$, we certainly want $a = b$. In other words, an ordering should be antisymmetric.

For example, over the natural numbers we recognize that the relation $<$ is an ordering and we notice that it is transitive and antisymmetric. Similarly, the relation $\leq$ is an ordering and we notice that it is transitive and antisymmetric. So the two essential properties of any kind of order are antisymmetric and transitive.

Let's look at how different orderings can occur in trying to perform the tasks of a recipe.

example 4.30 A Pancake Recipe

Suppose we have the following recipe for making pancakes.

1. Mix the dry ingredients (flour, sugar, baking powder) in a bowl.

2. Mix the wet ingredients (milk, eggs) in a bowl.

3. Mix the wet and dry ingredients together.

4. Oil the pan. (It's an old pan.)

5. Heat the pan.

6. Make a test pancake and throw it away.

7. Make pancakes.

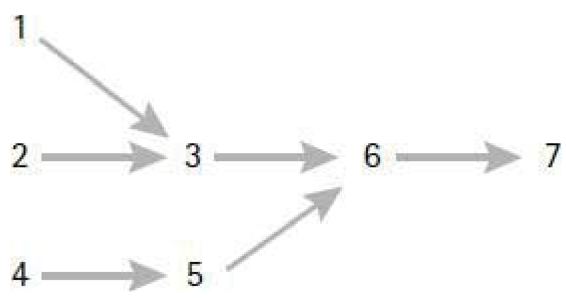


Figure 4.12 A pancake recipe.

Steps 1 through 7 indicate an ordering for the steps of the recipe. But the steps could also be done in some other order. To help us discover some other orders, let's define a relation R on the seven steps of the pancake recipe as follows:

$$i \, R \, j \text{ means that step } i \text{ must be done before step } j.$$

Notice that R is antisymmetric and transitive. We can picture R as the digraph (without the transitive arrows) in Figure 4.12.

The graph helps us pick out different orders for the steps of the recipe. For example, the following ordering of steps will produce pancakes just as well.

$$4, 5, 2, 1, 3, 6, 7.$$

So there are several ways to perform the recipe. For example, three people could work in parallel doing tasks 1, 2, and 4 at the same time.

end example

This example demonstrates that different orderings for time-oriented tasks are possible whenever some tasks can be done at different times without changing the outcome. The orderings can be discovered by modeling the tasks by a binary relation R defined by

$$i \, R \, j \text{ means that step } i \text{ must be done before step } j.$$

Notice that R is irreflexive because time-oriented tasks can't be done before themselves. If there are at least two tasks that are not related by R, as in Example 4.30, then there will be at least two different orderings of the tasks.

4.3.1 Partial Orders

Now let's get down to business and discuss the basic ideas and techniques of ordering. The two essential properties of order suffice to define the notion of partial order.

Definition of a Partial Order
A binary relation is called a *partial order* if it is antisymmetric, transitive, and either reflexive or irreflexive.

We should note that this definition is a bit more general than most definitions found in the literature. Some definitions require the reflexive property, while others require the irreflexive property. The reasons for requiring one of these properties are mostly historical and neither property is required to describe the idea of ordering. So if a partial order is reflexive and we wish to emphasize it, we'll call it a *reflexive partial order*. For example, $\leq$ is a reflexive partial order on the integers. If a partial order is irreflexive and we wish to emphasize it, we'll call it an *irreflexive partial order*. For example, $<$ is an irreflexive partial order on the integers.

Definition of a Partially Ordered Set
The set over which a partial order is defined is called a *partially ordered set*—or *poset* for short. If we want to emphasize the fact that R is the partial order that makes S a poset, we'll write $\langle S, R\rangle$ and call it a poset.

For example, in our pancake example we defined a partial order R on the set of recipe steps $\{1, 2, 3, 4, 5, 6, 7\}$. So we can say that $\langle\{1, 2, 3, 4, 5, 6, 7\}, R\rangle$ is a poset. There are many more examples of partial orders. For example, $\langle \mathbb{N}, <\rangle$ and $\langle \mathbb{N}, \leq\rangle$ are posets because the relations $<$ and $\leq$ are both antisymmetric and transitive.

Partial and Total

The word "partial" is used in the definition because we include the possibility that some elements may not be related to each other, as in the pancake recipe example. For another example, consider the subset relation on power($\{a, b, c\}$). Certainly the subset relation is antisymmetric and transitive. So we can say that $\langle$power($\{a, b, c\}$), $\subseteq$ is a poset. Notice that there are some subsets that are not related. For example, $\{a, b\}$ and $\{a, c\}$ are not related by the relation $\subseteq$.

Suppose R is a binary relation on a set S and $x, y \in S$. We say that x and y are *comparable* if either $x \; R \; y$ or $y \; R \; x$. In other words, elements that are related are comparable. If every pair of distinct elements in a partial order are comparable, then the order is called a *total* order (also called a *linear* order). If R is a total order on the set S, then we also say that S is a *totally ordered set* or a *linearly ordered set*. For example, the natural numbers are totally ordered by both "less" and "lessOrEqual." In other words, $\mathbb{N}, <$ and $\mathbb{N}, \leq$ are totally ordered sets.

example 4.31 The Divides Relation

Let's look at some interesting posets that can be defined by the divides relation, |. First we'll consider the set $\mathbb{N}$. If $a|b$ and $b|c$, then $a|c$. Thus | is transitive. Also, if $a|b$ and $b|a$, then it must be the case that $a = b$. So | is antisymmetric. Therefore,

$$\mathbb{N}, | \text{ is a poset.}$$

But $\mathbb{N}, |$ is not totally ordered because, for example, 2 and 3 are not comparable. To obtain a total order, we need to consider subsets of $\mathbb{N}$. For example, it's easy to see that for any m and n, either $2^m|2^n$ or $2^n|2^m$. Therefore,

$$\{2^n \mid n \in \mathbb{N}\}, | \text{ is a totally ordered set.}$$

Let's consider some finite subsets of $\mathbb{N}$. For example, it's easy to see that

$$\{1, 3, 9, 45\}, | \text{ is a totally ordered set.}$$

It's also easy to see that

$$\langle\{1, 2, 3, 4\}, |\rangle \text{ is a poset that is not totally ordered}$$

because 3 can't be compared to either 2 or 4.

end example

Notation for Partial Orders

When talking about partial orders, we'll often use the symbols

$$\prec \text{ and } \preceq$$

to stand for an irreflexive partial order and a reflexive partial order, respectively. We can read $a \prec b$ as "a is less than b," and we can read $a \preceq b$ as "a is less than or equal to b." The two symbols can be defined in terms of each other. For example, if $\langle A, \prec\rangle$ is a poset, then we can define the relation $\preceq$ in terms of $\prec$ by writing

$$\preceq \; = \; \prec \cup \{(x, x) \mid x \in A\}.$$

In other words, $\preceq$ is the reflexive closure of $\prec$. So $x \preceq y$ always means $x \prec y$ or $x = y$. Similarly, if $\langle B, \preceq\rangle$ is a poset, then we can define the relation $\prec$ in terms of $\preceq$ by writing

$$\prec \; = \; \preceq - \{(x, x) \mid x \in B\}.$$

Therefore, $x \prec y$ always means $x \preceq y$ and $x \neq y$. We also write the expression $y \succ x$ to mean the same thing as $x \prec y$.

Chains

A set of elements in a poset is called a *chain* if all the elements are comparable—linked—to each other. For example, any totally ordered set is itself a chain. A sequence of elements $x_1, x_2, x_3, \ldots$ in a poset is said to be a *descending chain* if $x_i \succ x_{i+1}$ for each $i \geq 1$. We can write the descending chain in the following familiar form:

$$x_1 \succ x_2 \succ x_3 \succ \cdots.$$

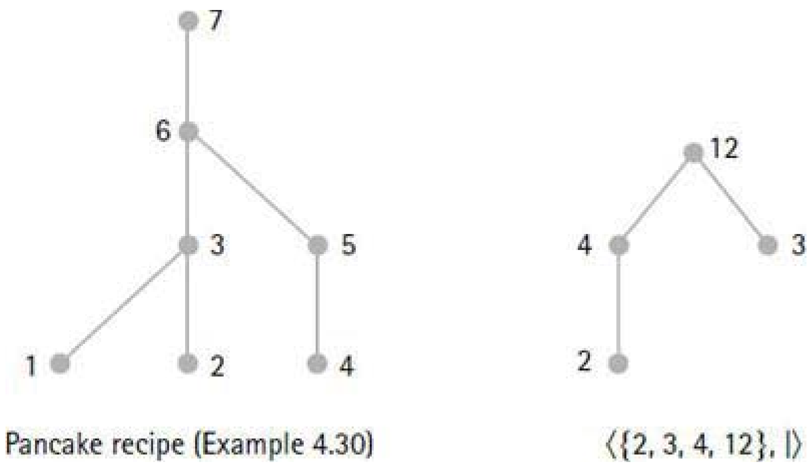


Figure 4.13 Two poset diagrams.

For example, $4 > 2 > 0 > -2 > -4 > -6 > \ldots$ is a descending chain in $\langle \mathbb{Z}, < \rangle$. For another example, $\{a, b, c\} \supset \{a, b\} \supset \{a\} \supset \varnothing$ is a finite descending chain in $\langle \text{power}(\{a, b, c\}), \subseteq \rangle$. We can define an *ascending chain* of elements in a similar way. For example, $1 \mid 2 \mid 4 \mid \ldots \mid 2^n \mid \ldots$ is an ascending chain in the poset $\langle \mathbb{N}, \mid \rangle$.

Predecessors and Successors

If $x \prec y$, then we say that x is a *predecessor* of y, or y is a *successor* of x. Suppose that $x \prec y$ and there are no elements between x and y. In other words, suppose we have the following situation:

$$\{z \in A \mid x \prec z \prec y\} = \varnothing.$$

When this is the case, we say that x is an *immediate predecessor* of y, or y is an *immediate successor* of x. In a finite poset an element with a successor has an immediate successor. Some infinite posets also have this property. For example, every natural number x has an immediate successor $x + 1$ with respect to the "less" relation. But no rational number has an immediate successor with respect to the "less" relation.

Poset Diagrams

A poset can be represented by a special graph called a *poset diagram* or a *Hasse diagram*—after the mathematician Helmut Hasse (1898–1979).

Whenever $x \prec y$ and x is an immediate predecessor of y, then place an edge (x, y) in the poset diagram with x at a lower level than y. A poset diagram can often help us observe certain properties of a poset. For example, the two poset diagrams in Figure 4.13 represent the pancake recipe poset from Example 4.30 and the poset $\langle\{2, 3, 4, 12\}, |\rangle$.

The three poset diagrams shown in Figure 4.14 are for the natural numbers and the integers with their usual orderings and for power($\{a, b\}$) with the subset relation.

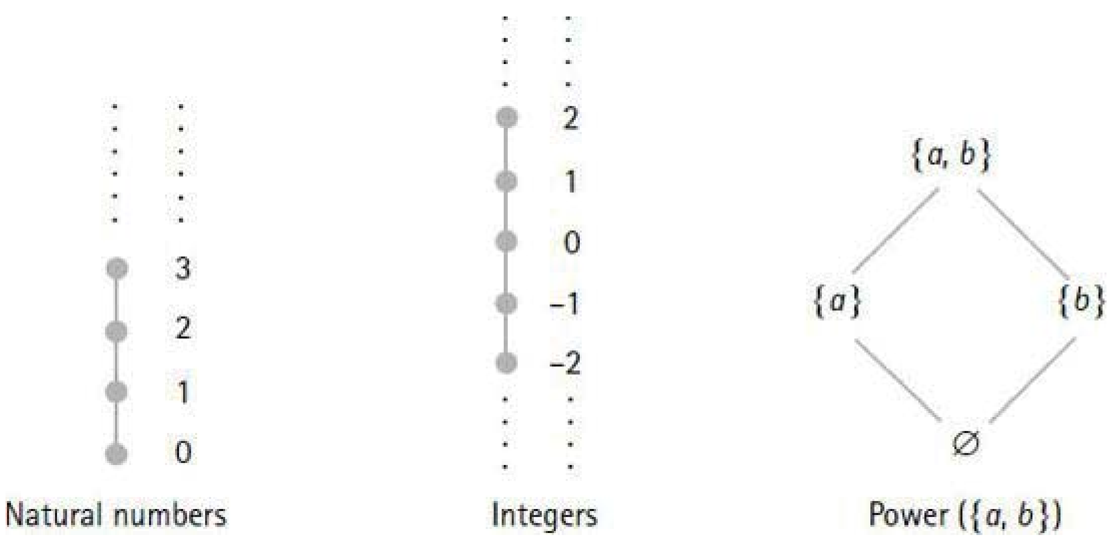


Figure 4.14 Three poset diagrams.

Maxima, Minima, and Bounds

When we have a partially ordered set, it's natural to use words like "minimal," "least," "maximal," and "greatest." Let's give these words some formal definitions.

Suppose that S is any nonempty subset of a poset P. An element $x \in S$ is called a *minimal element* of S if x has no predecessors in S. An element $x \in S$ is called the *least element* of S if $x \preceq y$ for all $y \in S$. For example, let's consider the poset $\langle\mathbb{N}, |\rangle$.

The subset $\{2, 4, 5, 10\}$ has two minimal elements, 2 and 5.

The subset $\{2, 4, 12\}$ has least element 2.

The set $\mathbb{N}$ has least element 1 because $1|x$ for all $x \in \mathbb{N}$.

For another example, let's consider the poset $\langle \text{power}(\{a, b, c\}), \subseteq \rangle$. The subset $\{\{a, b\}, \{a\}, \{b\}\}$ has two minimal elements, $\{a\}$ and $\{b\}$. The power set itself has least element $\varnothing$.

In a similar way we can define *maximal elements* and the *greatest element* of a subset of a poset. For example, let's consider the poset $\langle \mathbb{N}, | \rangle$.

The subset $\{2, 4, 5, 10\}$ has two maximal elements, 4 and 10.

The subset $\{2, 4, 12\}$ has greatest element 12.

The set $\mathbb{N}$ itself has greatest element 0 because $x|0$ for all $x \in \mathbb{N}$.

For another example, let's consider the poset $\langle \text{power}(\{a, b, c\}), \subseteq \rangle$. The subset $\{\varnothing, \{a\}, \{b\}\}$ has two maximal elements, $\{a\}$ and $\{b\}$. The power set itself has greatest element $\{a, b, c\}$.

Some sets may not have any minimal elements, yet still be bounded below by some element. For example, the set of positive rational numbers has no least element yet is bounded below by the number 0. Let's introduce some standard terminology that can be used to discuss ideas like this.

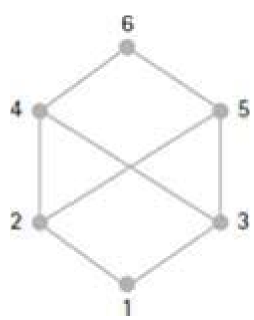


Figure 4.15 A poset diagram.

If S is a nonempty subset of a poset P, an element $x \in P$ is called a *lower bound* of S if $x \preceq y$ for all $y \in S$. An element $x \in P$ is called the *greatest lower bound* (or *glb*) of S if x is a lower bound and $z \preceq x$ for all lower bounds z of S. The expression glb(S) denotes the greatest lower bound of S, if it exists. For example, if we let $\mathbb{Q}^+$ denote the set of positive rational numbers, then over the poset $\langle \mathbb{Q}, \leq \rangle$ we have $\text{glb}(\mathbb{Q}^+) = 0$.

In a similar way we define upper bounds for a subset S of the poset P. An element $x \in P$ is called an *upper bound* of S if $y \preceq x$ for all $y \in S$. An element $x \in P$ is called the *least upper bound* (or *lub*) of S if x is an upper bound and x

$\preceq z$ for all upper bounds z of S. The expression lub(S) denotes the least upper bound of S, if it exists. For example, lub($\mathbb{Q}^+$) does not exist in $\langle \mathbb{Q}, \leq \rangle$.

For another example, in the poset $\langle$N, $\leq\rangle$, every finite subset has a glb—the least element—and a lub—the greatest element. Every infinite subset has a glb but no upper bound.

Can subsets have upper bounds without having a least upper bound? Sure. Here's an example.

example 4.32 Upper Bounds

Suppose the set {1, 2, 3, 4, 5, 6} represents six time-oriented tasks. You can think of the numbers as chapters in a book, as processes to be executed on a computer, or as the steps in a recipe for making ice cream. In any case, suppose the tasks are partially ordered according to the poset diagram in Figure 4.15.

The subset {2, 3} is bounded above, but it has no least upper bound. Notice that 4, 5, and 6 are all upper bounds of {2, 3}, but none of them is a least upper bound.

end example

Lattices

A *lattice* is a poset with the property that every pair of elements has a glb and a lub. So the poset of Example 4.32 is not a lattice. For example, $\langle \mathbb{N}, \leq \rangle$ is a lattice in which the glb of two elements is their minimum and the lub is their maximum. For another example, if A is any set, then $\langle$power(A), $\subseteq\rangle$ is a lattice, where glb(X, Y) = $X \cap Y$ and lub(X, Y)= $X \cup Y$. The word "lattice" is used because lattices that aren't totally ordered often have poset diagrams that look like "latticeworks" or "trellisworks."

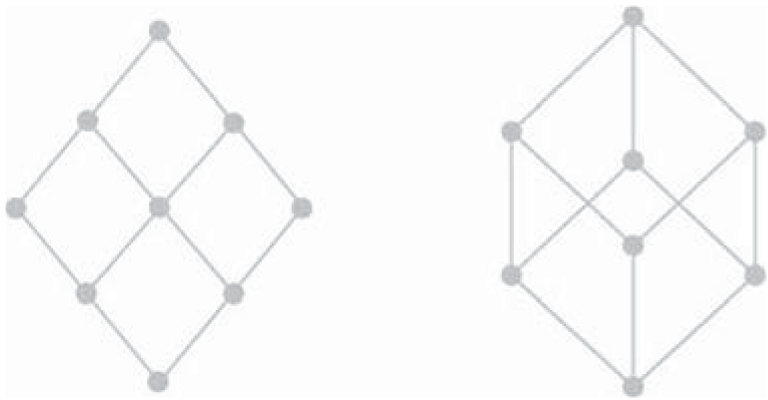


Figure 4.16 Two lattices.

For example, the two poset diagrams in Figure 4.16 represent lattices. These two poset diagrams can represent many different lattices. For example, the poset diagram on the left represents the lattice whose elements are the positive divisors of 36, ordered by the divides relation. In other words, it represents the lattice $\langle\{1, 2, 3, 4, 6, 9, 12, 18, 36\}, \mid$. See whether you can label the poset diagram with these numbers. The diagram on the right represents the lattice power($\{a, b, c\}$), $\subseteq$. It also represents the lattice whose elements are the positive divisors of 70, ordered by the divides relation. See whether you can label the poset diagram with both of these lattices. We'll give some more examples in the exercises.

4.3.2 Topological Sorting

A typical computing task is to sort a list of elements taken from a totally ordered set. Here's the problem statement.

> **The Sorting Problem**
> Find an algorithm to sort a list of elements from a totally ordered set.

For example, suppose we're given the list $x_1, x_2, \ldots, x_n$, where the elements of the list are related by a total order relation R. We might sort the list by a program "sort," which we could call as follows:

$$\text{sort}(R,\ x_1, x_2, \ldots, x_n\ ).$$

For example, we should be able to obtain the following results with sort:

$$\text{sort}\,(<, \langle 8, 3, 10, 5\rangle) = \langle 3, 5, 8, 10\rangle\,,$$
$$\text{sort}\,(>, \langle 8, 3, 10, 5\rangle) = \langle 10, 8, 5, 3\rangle\,.$$

Programming languages normally come equipped with several totally ordered sets. If a total order R is not part of the language, then R must be implemented as a relational test, which can then be called into action whenever a comparison is required in the sorting algorithm.

Topological Sorting

Can a partially ordered set be sorted? The answer is yes if we broaden our idea of what sorting means. Here's the problem statement.

The Topological Sorting Problem
Find an algorithm to sort a list of elements from a partially ordered set.

How can we "sort" a list when some elements may not be comparable? Well, we try to find a listing that maintains the partial ordering, as in the pancake recipe from Example 4.30. Given a partial order R on a set, a list of elements from the set is *topologically sorted* if whenever two elements in the list satisfy $a\ R\ b$, then a is to the left of b in the list.

The ordering of a set of tasks is a topological sorting problem. For example, the list $\langle 4, 5, 2, 1, 3, 6, 7\rangle$ is a topological sort of the steps in the pancake recipe from Example 4.30. Another example of a topological sort is the ordering of the chapters in a textbook in which the partial order is defined to be the dependence of one chapter upon another. In other words, we hope that we don't have to read some chapter further on in the book to understand what we're reading now.

Is there a technique to do topological sorting? Yes. Suppose R is a partial order on a finite set A. For each element $y \in A$, let $P\ (y)$ be the number of immediate predecessors of y, and let $S(y)$ be the set of immediate successors of y. Let *Sources* be the set of sources—minimal elements—in A. Therefore, y

is a source if and only if $P(y) = 0$. A topological sort algorithm goes something like the following:

Topological Sort Algorithm **(4.17)**

While the set of sources is not empty, do the following steps:

1. Output a source y.

2. For all z in $S(y)$, decrement $P(z)$; if $P(z) = 0$, then add z to *Sources*.

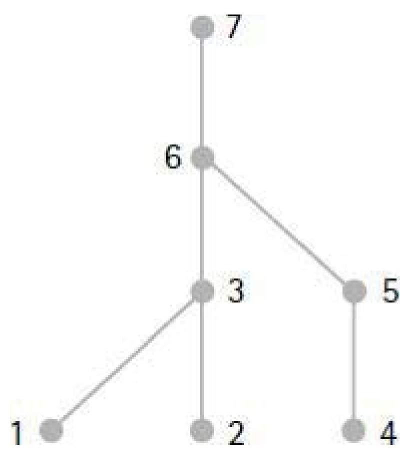


Figure 4.17 Poset of a pancake recipe.

A more detailed description of the algorithm can be given as follows:

Detailed Topological Sort

while $Sources \neq \varnothing$ **do**
  Pick a source y from *Sources;*
  Output y;
  for each z in $S(y)$ **do**
    $P(z) := P(z) - 1$;
    if $P(z) = 0$ **then** $Sources := Sources \cup \{z\}$
  od;
  $Sources := Sources - \{y\}$;
od

Let's do an example that includes some details on how the data for the algorithm might be represented.

example 4.33 A Topological Sort

We'll consider the steps of the pancake recipe from Example 4.30. Figure 4.17 shows the poset diagram for the steps of the recipe.

The initial set of sources is $\{1, 2, 4\}$. Letting P be an array of integers, we get the following initial table of predecessor counts:

i	1	2	3	4	5	6	7
$P(i)$	0	0	2	0	1	2	1

The following table is the initial table of successor sets S:

i	1	2	3	4	5	6	7
$S(i)$	$\{3\}$	$\{3\}$	$\{6\}$	$\{5\}$	$\{6\}$	$\{7\}$	$\varnothing$

You should trace the algorithm for these data representations.

end example

There is a very interesting and efficient implementation of algorithm (4.17) in Knuth [1968]. It involves the construction of a novel data structure to represent the set of sources, the sets $S(y)$ for each y, and the numbers $P(z)$ for each z.

4.3.3 Well-Founded Orders

Let's look at a special property of the natural numbers. Suppose we're given a descending chain of natural numbers that begins as follows:

$$29 > 27 > 25 > \cdots.$$

Can this descending chain continue forever? Of course not. We know that 0 is the least natural number, so the given chain must stop after only a finite number of terms. This is not an earthshaking discovery, but it is an example of the property of well-foundedness that we're about to discuss.

Definition of a Well-Founded Order

We're going to consider posets with the property that every descending chain of elements is finite. So we'll give these posets a name.

> **Well-Founded**
>
> A poset is said to be *well-founded* if every descending chain of elements is finite. In this case, the partial order is called a *well-founded order*.

For example, we've seen that $\mathbb{N}$ is a well-founded set with respect to the less relation $<$. In fact, any set of integers with a least element is well-founded by $<$. For example, the following three sets of integers are well-founded.

$$\{1, 2, 3, 4, \ldots\}, \{m \mid m \geq -3\}, \text{ and } \{5, 9, 13, 17, \ldots\}.$$

For another example, any collection of finite sets is well-founded by $\subseteq$. This is easy to see because any descending chain must start with a finite set. If the set has n elements, it can start a descending chain of at most $n + 1$ subsets. For example, the following expression displays a longest descending chain starting with the set $\{a, b, c\}$.

$$\{a, b, c\} \supset \{b, c\} \supset \{c\} \supset \varnothing.$$

So the power set of a finite set is well-founded with respect to $\subseteq$.

But many posets are not well-founded. For example, the integers and the positive rationals are not well-founded with respect to the less relation because they have infinite descending chains as the following examples show.

$$2 > 0 > -2 > -4 > \cdots$$

$$\frac{1}{2} > \frac{1}{3} > \frac{1}{4} > \frac{1}{5} > \cdots .$$

The power set of an infinite set is not well-founded by $\subset$. For example, if we let $S_k = \mathbb{N} - \{0, 1, \ldots, k\}$, then we obtain the following infinite descending chain in power($\mathbb{N}$):

$$S_0 \supset S_1 \supset S_2 \supset \cdots \supset S_k \supset \cdots.$$

Are well-founded sets good for anything? The answer is yes. We'll see in the next section that they are basic tools for inductive proofs. So we should get familiar with them. We'll do this by looking at another property that well-founded sets possess.

The Minimal Element Property

Does every subset of $\mathbb{N}$ have a least element? A quick-witted person might say, "Yes," and then think a minute and say, "except that the empty set doesn't have any elements, so it can't have a least element." Suppose the question is modified to "Does every nonempty subset of $\mathbb{N}$ have a least element?" Then a bit of thought will convince most of us that the answer is yes.

We might reason as follows: Suppose S is some nonempty subset of $\mathbb{N}$ and x_1 is some element of S. If x_1 is the least element of S, then we are done. So assume that x_1 is not the least element of S. Then x_1 must have a predecessor x_2 in S—otherwise, x_1 would be the least element of S. If x_2 is the least element of S, then we are done. If x_2 is not the least element of S, then it has a predecessor x_3 in S, and so on. If we continue in this manner, we will obtain a descending chain of distinct elements in S:

$$x_1 > x_2 > x_3 > \ldots .$$

This looks familiar. We already know that this chain of natural numbers can't be infinite. So it stops at some value, which must be the least element of S. So every nonempty subset of the natural numbers has a least element.

This property is not true for all posets. For example, the set of integers has no least element. The open interval of real numbers (0, 1) has no least element. Also the power set of a finite set can have collections of subsets that have no least element.

Notice however that every collection of subsets of a finite set does contain a minimal element. For example, the collection $\{\{a\}, \{b\}, \{a, b\}\}$ has two minimal elements $\{a\}$ and $\{b\}$. Remember, the property that we are looking for must be true for all well-founded sets. So the existence of least elements is out; it's too restrictive.

But what about the existence of minimal elements for nonempty subsets of a well-founded set? This property is true for the natural numbers. (Least elements are certainly minimal.) It's also true for power sets of finite sets. In fact, this property is true for all well-founded sets, and we can state the result as follows:

Descending Chains and Minimality (4.18)

If A is a well-founded set, then every nonempty subset of A has a minimal element. Conversely, if every nonempty subset of A has a minimal element, then A is well-founded.

It follows from (4.18) that the property of finite descending chains is equivalent to the property of nonempty subsets having minimal elements. In other words, if a poset has one of the properties, then it also has the other property. Thus it is also correct to define a well-founded set to be a poset with the property that every nonempty subset has a minimal element. We will call this latter property the *minimum condition* on a poset.[1]

Whenever a well-founded set is totally ordered, then each nonempty subset has a single minimal element, the least element. Such a set is called a *well-ordered set*. So a well-ordered set is a totally ordered set such that every nonempty subset has a least element. For example, $\mathbb{N}$ is well-ordered by the "less" relation. Let's examine a few more total orderings to see whether they are well-ordered.

Lexicographic Ordering of Tuples

The linear ordering $<$ on $\mathbb{N}$ can be used to create the *lexicographic* order on

$\mathbb{N}^k$, which is defined as follows.

$$(x_1, \ldots, x_k) \prec (y_1, \ldots, y_k)$$

if and only if there is an index $j \geq 1$ such that $x_j < y_j$ and for each $i < j$, $x_i = y_i$. This ordering is a total ordering on $\mathbb{N}^k$. It's also a well-ordering.

For example, the lexicographic order on $\mathbb{N} \times \mathbb{N}$ has least element (0, 0). Every nonempty subset of $\mathbb{N} \times \mathbb{N}$ has a least element, namely, the pair (x, y) with the smallest value of x, where y is the smallest value among second components of pairs with x as the first component. For example, (0, 10) is the least element in the set {(0, 10), (0, 11), (1, 0)}. Notice that (1, 0) has infinitely many predecessors of the form $(0, y)$, but (1, 0) has no immediate predecessor.

Lexicographic Ordering of Strings

Another type of lexicographic ordering involves strings. To describe it we need to define the prefix of a string. If a string x can be written as $x = uv$ for some strings u and v, then u is called a *prefix* of x. If $v \neq \Lambda$, then u is a *proper prefix* of x. For example, the prefixes of the string *aba* over the alphabet $\{a, b\}$ are Λ, *a, ab*, and *aba*. The proper prefixes of *aba* are Λ, *a*, and *ab*.

Definition of Lexicographic Ordering

Let A be a finite alphabet with some agreed-upon linear ordering. Then the *lexicographic* ordering on A^* is defined as follows: $x \prec y$ iff either x is a proper prefix of y or x and y have a longest common proper prefix u such that $x = uv$, $y = uw$, and head(v) precedes head(w) in A.

The lexicographic ordering on A^* is often called the *dictionary ordering* because it corresponds to the ordering of words that occur in a dictionary. The definition tells us that if $x \neq y$, then either $x \prec y$ or $y \prec x$. So the lexicographic ordering on A^* is a total (i.e., linear) ordering. It also follows that every string x has an immediate successor xa, where a is the first letter of A.

If A has at least two elements, then the lexicographic ordering on A^* is *not* well-ordered. For example, let $A = \{a, b\}$ and suppose that a precedes b. Then the elements in the set $\{a^n b \mid n \in \mathbb{N}\}$ form an infinite descending chain:

$$b \succ ab \succ aab \succ aaab \succ \cdots \succ a^n b \succ \cdots .$$

Notice also that b has no immediate predecessor because if $x \prec b$, then we have $x \prec xa \prec b$.

Standard Ordering of Strings

Now let's look at an ordering that is well-ordered. The *standard ordering on strings* uses a combination of length and the lexicographic ordering.

> **Definition of Standard Ordering**
> Let A be a finite alphabet with some agreed-upon linear ordering. The standard ordering on A^* is defined as follows, where $\prec_L$ denotes the lexicographic ordering on A^*:
>
> $x \prec y$ iff either length(x) < length(y), or length(x) = length(y) and $x \prec_L y$.

It's easy to see that $\prec$ is a total order and every string has an immediate successor and an immediate predecessor. The standard ordering on A^* is also well-ordered because each string has a finite number of predecessors. For example, let $A = \{a, b\}$ and suppose that a precedes b. Then the first few elements in the standard order of A^* are given as follows:

$$\Lambda, a, b, aa, ab, ba, bb, aaa, aab, aba, abb, baa, bab, bba, bbb, \ldots .$$

Constructing Well-Founded Orderings

Collections of strings, lists, trees, graphs, or other structures that programs process can usually be made into well-founded sets by defining an appropriate order relation. For example, any finite set can be made into a well-founded set—actually a well-ordered set—by simply listing its elements in any order we wish, letting the leftmost element be the least element.

Let's look at some ways to build well-founded orderings for infinite sets. Suppose we want to define a well-founded order on some infinite set S. A simple and useful technique is to associate each element of S with some element in an existing well-founded set. For example, the natural numbers are well-founded by $<$. So we'll use them as a building block for well-founded constructions.

Constructing a Well-Founded Order (4.19)

Given any function $f : S \to \mathbb{N}$, there is a well-founded order $\prec$ defined on S in the following way, where $x, y \in S$:

$$x \prec y \text{ means } f(x) < f(y).$$

Does the new relation $\prec$ make S into a well-founded set? Sure. Suppose we have a descending chain of elements in S as follows:

$$x_1 \succ x_2 \succ x_3 \succ \cdots.$$

The chain must stop because $x \succ y$ is defined to mean $f(x) > f(y)$, and we know that any descending chain of natural numbers must stop. Let's look at a few more examples.

example 4.34 Some Well-Founded Orderings

a. Any set of lists, where $L \prec M$ means length(L) < length(M).

b. Any set of strings, where $s \prec t$ means length(s) < length(t).

c. Any set of trees, where $B \prec C$ means the number of nodes in B is less than the number of nodes in C.

d. Any set of trees, where $B \prec C$ means the number of leaves in B is less than the number of leaves in C.

e. Any set of nonempty trees, where $B \prec C$ means depth(B) < depth(C).

f. Any set of people can be well-founded by the age at the last birthday of each person. What are the minimal elements?

g. The set $\{\ldots, -3, -2, -1\}$ of negative integers, where $x \prec y$ means $x > y$.

end example

As the examples show, we can use known properties of structures to find useful well-founded orderings for sets of structures. The next example constructs a finite, hence well-founded, lexicographic order.

example 4.35 A Finite Lexicographic Order

Let $S = \{0, 1, 2, \ldots, m\}$. Then we can define a lexicographic ordering on the set S^k in a natural way. Since S is finite, it follows that the lexicographic ordering on S^k is well-founded. The least element is $(0, \ldots, 0)$, and the greatest element is $(m, \ldots, m)$. For example, if $k = 3$, then the immediate successor of any element can be defined as

$$\begin{aligned}\operatorname{succ}((x, y, z)) = {} & \text{if } z < m \text{ then } (x, y, z+1)\\ & \text{else if } y < m \text{ then } (x, y+1, 0)\\ & \text{else if } x < m \text{ then } (x+1, 0, 0)\\ & \text{else error (no successor).}\end{aligned}$$

end example

Inductively Defined Sets are Well-Founded

It's easy to make an inductively defined set W into a well-founded set if no two elements are defined in terms of each other. We'll give two methods. Both methods let the basis elements of W be the minimal elements of the well-founded order.

Method 1: (4.20)

Define a function $f: W \to \mathbb{N}$ as follows:

1. $f(c) = 0$ for all basis elements c of W.

2. If $x \in W$ and x is constructed from elements $y_1, y_2, \ldots, y_n$ in W, then define $f(x) = 1 +$

$\max\{f(y_1), f(y_2), \ldots, f(y_n)\}$.

Let $x \prec y$ mean $f(x) < f(y)$.

Since 0 is the least element of $\mathbb{N}$ and $f(c) = 0$ for all basis elements c of W, it follows that the basis elements of W are minimal elements under the ordering defined by (4.20). For example, if c is a basis element of W and if $x \prec c$, then $f(x) < f(c) = 0$, which can't happen with natural numbers. Therefore c is a minimal element of W.

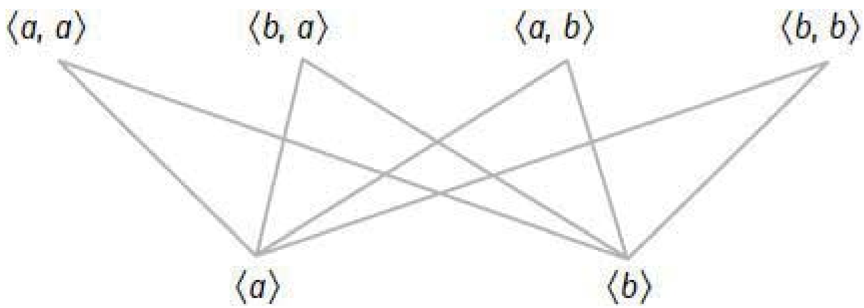


Figure 4.18 Part of a poset diagram.

example 4.36 A Well-Founded Ordering

Let W be the set of all nonempty lists over $\{a, b\}$. First we'll give an inductive definition of W. The lists ⟨a and b are the basis elements of W. For the induction case, if $L \in W$, then the lists cons(a, L) and cons(b, L) are in W. Now we'll use (4.20) to make W into a well-founded set. The function f of (4.20) turns out to be $f(L) = \text{length}(L) - 1$. So for any lists L and M in W, we define $L \prec M$ to mean $f(L) < f(M)$, which means $\text{length}(L) - 1 < \text{length}(M) - 1$, which also means $\text{length}(L) < \text{length}(M)$. The diagram in Figure 4.18 shows the bottom two layers of a poset diagram for W with its two minimal lists a and b .

end example

Method 1 can relate many elements. For example, to add one more level to the diagram in Figure 4.18, we have to include eight 3-element lists and draw 32 lines from the two element lists up to the three element lists. Sometimes it isn't necessary to have an ordering that relates so many elements. This brings us to the second method for defining a well-founded ordering on an inductively defined set W.

> *Method 2:* (4.21)
>
> The ordering $\prec$ is defined as follows:
>
> **1.** Let the basis elements of W be minimal elements.
>
> **2.** If $x \in W$ and x is constructed from elements $y_1, y_2, \ldots, y_n$ in W, then define $y_i \prec x$ for each $i = 1, \ldots, n$.
>
> The actual ordering is the transitive closure of $\prec$.

The ordering of (4.21) is well-founded because any x can be constructed from basis elements with finitely many constructions. Therefore, there can be no infinite descending chain starting at x. With this ordering, there can be many pairs that are not related.

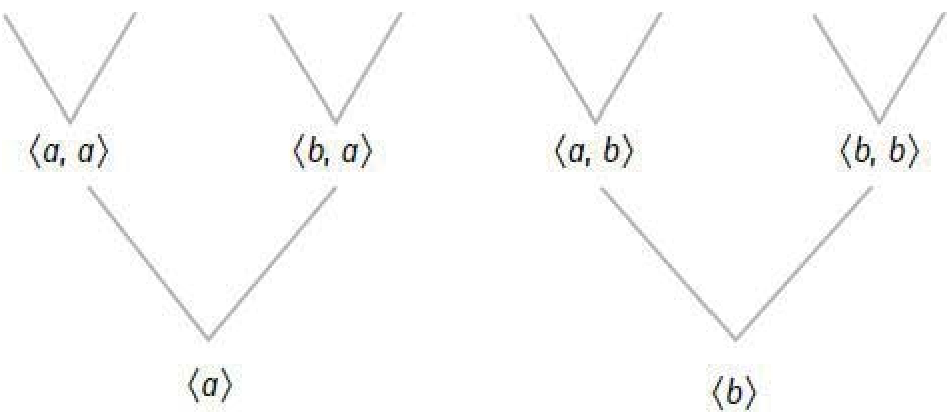


Figure 4.19 Part of a poset diagram.

For example, we'll use the preceding example of nonempty lists over the set $\{a, b\}$. The picture in Figure 4.19 shows the bottom two levels of the poset diagram for the well-founded ordering constructed by (4.21).

Notice that each list has only two immediate successors. For example, the two successors of $\langle a \rangle$ are cons(a, $\langle a \rangle$) = $\langle a, a \rangle$ and cons(b, $\langle a \rangle$) = $\langle b, a \rangle$. The two successors of $\langle b, a \rangle$ are $\langle a, b, a \rangle$ and $\langle b, b, a \rangle$. This is much simpler than the ordering we got using (4.20).

Let's look at some examples of inductively defined sets that are well-founded sets by the method of (4.21).

example 4.37 Using One Part of a Product

We'll define the set $\mathbb{N} \times \mathbb{N}$ inductively by using the first copy of $\mathbb{N}$. For the basis case we put $(0, n) \in \mathbb{N} \times \mathbb{N}$ for all $n \in \mathbb{N}$. For the induction case, whenever the pair $(m, n) \in \mathbb{N} \times \mathbb{N}$, we put $(m + 1, n) \in \mathbb{N} \times \mathbb{N}$. The relation on $\mathbb{N} \times \mathbb{N}$ induced by this inductive definition and (4.21) is not linearly ordered.

For example, (0, 0) and (0, 1) are not related because they are both basis elements. Notice that any pair (m, n) is the beginning of a descending chain containing at most $m + 1$ pairs. For example, the following chain is the longest descending chain that starts with (3, 17).

$$(3, 17), (2, 17), (1, 17), (0, 17).$$

end example

example 4.38 Using Both Parts of a Product

Let's define the set $\mathbb{N} \times \mathbb{N}$ inductively by using both copies of $\mathbb{N}$. The single basis element is (0, 0). For the induction case, if $(m, n) \in \mathbb{N} \times \mathbb{N}$, then put $(m + 1, n)$, $(m, n + 1) \in \mathbb{N} \times \mathbb{N}$. Notice that each pair with both components nonzero is defined twice by this definition. The relation induced by this definition and (4.21) is nonlinear.

For example, the two pairs (2, 1) and (1, 2) are not related. Any pair (m, n) is the beginning of a descending chain of at most $m + n + 1$ pairs. For example, the following descending chain has maximum length among the descending chains that start at the pair (2, 3).

$$(2, 3), (2, 2), (1, 2), (1, 1), (0, 1), (0, 0).$$

Can you find a different chain of the same length starting at (2, 3)?

end example

4.3.4 Ordinal Numbers

We'll finish our discussion of order by introducing the *ordinal numbers*. These numbers are ordered, and they can be used to count things. An ordinal number may be represented as a set with certain properties. For example, any ordinal number x has an immediate successor defined by $\text{succ}(x) = x \cup \{x\}$. The expression $x + 1$ is also used to denote $\text{succ}(x)$. The natural numbers denote ordinal numbers when we define $0 = \varnothing$ and interpret + as addition, in which case it's easy to see that

$$x + 1 = \{0, \ldots, x\}.$$

For example, $1 = \{0\}$, $2 = \{0, 1\}$, and $5 = \{0, 1, 2, 3, 4\}$. In this way, each natural number is an ordinal number, called a *finite ordinal*.

Now let's define some *infinite ordinals*. The first infinite ordinal is

$$\omega = \{0, 1, 2, \ldots\},$$

the set of natural numbers. The next infinite ordinal is

$$\omega + 1 = \text{succ}(\omega) = \omega \cup \{\omega\} = \{\omega, 0, 1, \ldots\}.$$

If α is an ordinal number, we'll write $\alpha + n$ in place of $\text{succ}^n(\alpha)$. So the first four infinite ordinals are ω, $\omega + 1$, $\omega + 2$, and $\omega + 3$. The infinite ordinals continue in this fashion. To get beyond this sequence of ordinals, we need to make a definition similar to the one for ω. The main idea is that any ordinal number is the union of all its predecessors. For example, we define $\omega 2 = \omega \cup \{\omega, \omega + 1, \ldots\}$. The ordinals continue with $\omega 2 + 1, \omega 2 + 2$, and so on. Of course, we can continue and define $\omega 3 = \omega 2 \cup \{\omega 2, \omega 2 + 1, \ldots\}$. After $\omega, \omega 2, \omega 3, \ldots$ comes the ordinal ω^2. Then we get $\omega^2 + 1, \omega^2 + 2, \ldots$, and we eventually get $\omega^2 + \omega$. Of course, the process goes on forever.

We can order the ordinal numbers by defining $\alpha < \beta$ iff $\alpha \in \beta$. For example, we have $x < x + 1$ for any ordinal x because $x \in \text{succ}(x) = x + 1$. So we get the familiar ordering $0 < 1 < 2 < \ldots$ for the finite ordinals. For any finite ordinal n we have $n < \omega$ because $n \in \omega$. Similarly, we have $\omega < \omega + 1$, and for any finite ordinal n we have $\omega + n < \omega 2$. So it goes. There are also uncountable ordinals, the least of which is denoted by Ω. And the ordinals continue on after this too.

Although every ordinal number has an immediate successor, there are some ordinals that don't have any immediate predecessors. These ordinals are called *limit ordinals* because they are defined as "limits" or unions of all their predecessors. The limit ordinals that we've seen are

$$0,\ \omega,\ \omega 2,\ \omega 3, \ldots,\ \omega^2, \ldots,\ \Omega, \ldots.$$

An interesting fact about ordinal numbers states that for any set S there is a bijection between S and some ordinal number. For example, there is a bijection between the set $\{a, b, c\}$ and the ordinal number $3 = \{0, 1, 2\}$. For another example there are bijections between the set $\mathbb{N}$ of natural numbers and each of the ordinals $\omega, \omega + 1, \omega + 2, \ldots$. Some people define the cardinality of a set to be the least ordinal number that is bijective to the set. So we have $|\{a, b, c\}| = 3$ and $|\mathbb{N}| = \omega$.

More information about ordinal numbers—including ordinal arithmetic—can be found in the excellent book by Halmos [1960].

Exercises

Partial Orders

1. Sometimes our intuition about a symbol can be challenged. For example, suppose we define the relation $\prec$ on the integers by saying that $x \prec y$ means $|x| < |y|$. Assign the value True or False to each of the following statements.

  a. $-7 \prec 7$.
  b. $-7 \prec -6$.
  c. $6 \prec -7$.
  d. $-6 \prec 2$.

2. State whether each of the following relations is a partial order.

  a. isFatherOf.
  b. isAncestorOf.

c. isOlderThan.
d. isSisterOf.
e. $\{(a, b), (a, a), (b, a)\}$.
f. $\{(2, 1), (1, 3), (2, 3)\}$.

3. Draw a poset diagram for each of the following partially ordered relations.

a. $\{(a, a), (a, b), (b, c), (a, c), (a, d)\}$.
b. power($\{a, b, c\}$), with the subset relation.
c. lists($\{a, b\}$), where $L \prec M$ if length(L) < length(M).
d. The set of all binary trees over the set $\{a, b\}$ that contain either one or two nodes. Let $s \prec t$ mean that s is either the left or right subtree of t.

4. Suppose we wish to evaluate the following expression as a set of time-oriented tasks:

$$(f(x) + g(x))(f(x)g(x)).$$

We'll order the subexpressions by data dependency. In other words, an expression can't be evaluated until its data are available. So the subexpressions that occur in the evaluation process are

x, $f(x)$, $g(x)$, $f(x) + g(x)$, $f(x)g(x)$, and $(f(x) + g(x))(f(x)g(x))$.

Draw the poset diagram for the set of subexpressions. Is the poset a lattice?

5. For any positive integer n, let D_n be the set of positive divisors of n. The poset $\langle D_n, |$ is a lattice. Describe the glb and lub for any pair of elements.

Well-Founded Property

6. Why is it true that every partially ordered relation over a finite set is well-founded?

7. For each set S, show that the given partial order on S is well-founded.

a. Let S be a set of trees. Let $s \prec t$ mean that s has fewer nodes than t.

b. Let S be a set of trees. Let $s \prec t$ mean that s has fewer leaves than t.

c. Let S be a set of lists. Let $L \prec M$ mean that length(L) < length(M).

8. Example 4.38 discussed a well-founded ordering for the set $\mathbb{N} \times \mathbb{N}$. Use this ordering to construct two distinct descending chains that start at the pair (4, 3), both of which have maximum length.

9. Suppose we define the relation $\prec$ on $\mathbb{N} \times \mathbb{N}$ as follows:

$$(a, b) \prec (c, d) \quad \text{if and only if} \quad \max\{a, b\} < \max\{c, d\}.$$

Is $\mathbb{N} \times \mathbb{N}$ well-founded with respect to $\prec$?

Topological Sorting

10. Trace the topological sort algorithm (4.17) for the pancake recipe in Example 4.30 by starting with the source 1. There are several possible answers because any source can be output by the algorithm.

11. Describe a way to perform a topological sort that uses an adjacency matrix to represent the partial order.

Proofs and Challenges

12. Show that the two properties irreflexive and transitive imply the antisymmetric property. So an irreflexive partial order can be defined by just the two properties irreflexive and transitive.

13. Prove the two statements of (4.18).

14. For a poset P, a function $f : P \to P$ is said to be *monotonic* if $x \preceq y$ implies $f(x) \preceq f(y)$ for all $x, y \in P$. For each poset and function definition, determine whether the function is monotonic.

a. $\langle \mathbb{N}, < \rangle$, $f(x) = 2x + 3$.

b. $\langle \mathbb{N}, < \rangle$, $f(x) = x^2$.